Words of Concern

Point to be As Allan Bloom has said "Education is the movement from darkness to light". Through this handbook, I have tried to illuminate what might otherwise appear as black boxes to some. In doing so, I have used references from several other authors to synthesize or simplify or elaborate information. This is not possible without omitting details that I deem trivial while dilating the data that I consider relevant to topic. Every effort has been made to avoid errors. In spite of this, some errors might have crept in. Any errors or discrepancies noted maybe brought to my notice which I shall rectify in my next revision.

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Thanks,
BCA V Sem students.
Interface College of Computer Applications (ICCA)
Davangere



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Syllabus:

Course Code: DSE-E1
Course Title: Cloud Computing
Credits - 03
Sem - V
Hours- 42
Formative Assessment Marks: 40
Summative Assessment Marks: 60
Duration of SEA: 02 hrs.

Course Outcomes

After the successful completion of the course, the student will be able to:

1. Explain the core concepts of the cloud computing paradigm such as how And why this paradigm shift came about, the characteristics, advantages And challenges brought about by the various models and services in Cloud computing.
2. Apply the fundamental concepts in data centers to understand the trade-Offs in power, efficiency and cost.
3. Identify resource management fundamentals like resource abstraction, Sharing and sandboxing and outline their role in managing infrastructure In cloud computing.
4. Analyze various cloud programming models and apply them to slove problems On the cloud.

Unit I 8 Hrs

Introduction :

Different Computing Paradigms-Parallel Computing, Distributed Computing, Cluster Computing, Grid Computing, Cloud Computing etc., Comparison of various Computing Technologies; Cloud Computing Basics-What is Cloud Computing? History, Characteristic Features, Advantages Disadvantages, and Applications of Cloud Computing; Trends in Cloud Computing; Leading Cloud Platform Service Providers.

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Unit II 10 Hrs

Cloud Architecture:

Cloud Service Model-Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS), Comparision of different Service Models; Cloud Deployment Models- Public Cloud; Private Cloud, Hybrid Cloud, Community Cloud; Cloud Computing Architecture- Layered Architecture of Cloud. Virtualization-Definition, Features of virtualization; Types of Virtualizations- Hardware Virtualization, Server Virtualization, Application Virtualization, Storage Virtualization, Operating System Virtualization; Virtualization and Cloud Computing, Pros and Cons of Virtualization, Technology Examples- Xen: Paravirtualization, VMware: Full Virtualization, Microsoft Hyper-V.

Unit III 8 Hrs

Cloud Application Programming and the Aneka Platform:

Aneka Cloud Application Platform- Framework Overview, Anatomy of the Aneka

Container; Building Aneka Clouds(Infrastructure Organization, Logical Organization, Private Cloud Deployment Mode,Public Cloud Deployment Mode, Hybrid Cloud Deployment Mode); Cloud Programming and Management- Aneka SDK(Application Model and Service Model);
Management Tools(Infrastructure, Platforms and Application management).

Unit IV 8 Hrs

Cloud Platforms in Industry:
Amazon Web Services- Compute Services, Storage Services, Communication Services, Additional Services; Google AppEngine- Architecture and Core Concepts, Application Life-Cycle, Cost Model, Observations;
Microsoft Azure- Azure Core Concepts(Compute, Storage, Core Infrastructure and Other Services), SOL Azure, Windows Azure Platform Appliance.

Unit V 8 Hrs

Cloud Applications:
Scientific Applications- Healthcare(ECG Analysis in the Cloud)
Biology (Protein Structure Predication and Gene Expression Data Analysis for Cancer Diagnosis), Geoscience (Satellite Image Processing); and Consumer Applications- CRM and ERP, Productivity, Social Networking, Media Applications, Multiplayer Online Gaming.

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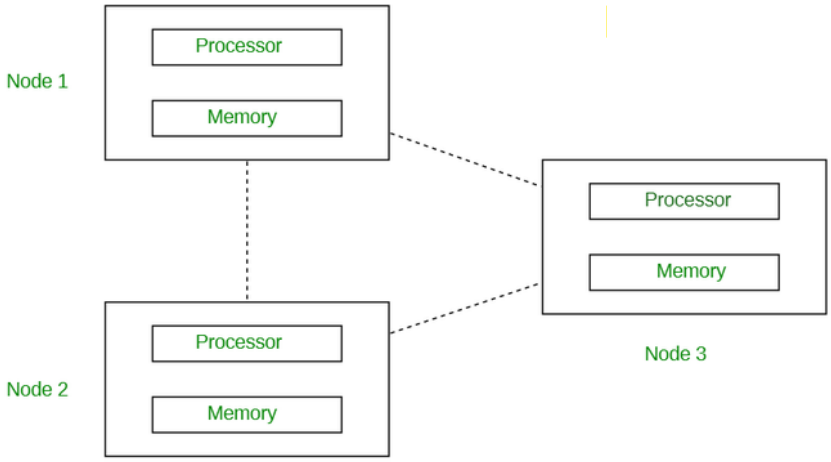
Unit I: Introduction

Computing

Computing refers to the process of using computers or computing devices to perform various takes, solve problems and manipulate data.

Types Of Computing Paradigms:

1. Distributed Computing
   - Distributed computing is defined as a type of computing where multiple computer systems work on a single problem. Here all the computer systems are linked together and the problem is divided into sub-problems where each part is solved by different computer systems.
   - The goal of distributed computing is to increase the performance and efficiency of the system and ensure fault tolerance.



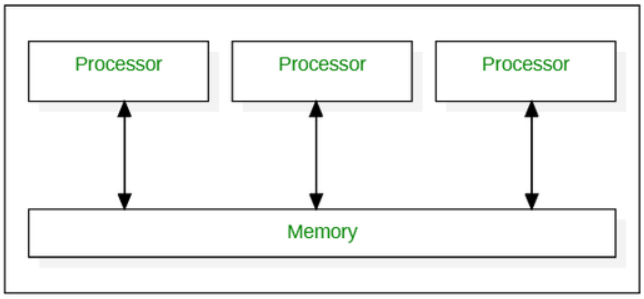
Distributed computing is used in a wide range of applications

- Web hosting: Distributed computing is used to host websites and other web applications. This allows websites to be accessed by users all over the world without having to rely on a single server.
- Email: Distributed computing is used to send and receive email messages. This allows email messages to be delivered quickly and reliably to users all over the world.
- Data storage: Distributed computing is used to store large amounts of data. This allows data to be accessed by users from anywhere in the world, and it also makes it easier to back up data and protect it from loss.

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2. Parallel Computing

   - Parallel computing refers to the process of breaking down larger problems into smaller, independent, often similar parts that can be executed simultaneously by multiple processors communicating via shared memory, the results of which are combined upon completion as part of an overall algorithm.
   - The goal of parallel computing is to save time and provide concurrency.



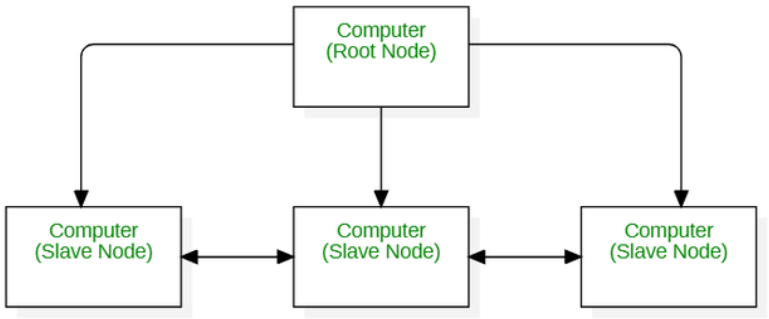
Parallel computing is used in a wide range of applications

- Scientific research: Parallel computing is used by scientists to solve complex problems that require a lot of computing power, such as weather forecasting, climate modelling, and drug discovery.
- Media and entertainment: Parallel computing is used to create and render high-quality graphics and videos, and to power special effects in movies and video games.
- Artificial intelligence: Parallel computing is used to train and deploy machine learning models, which are used for a variety of tasks, such as image recognition, natural language processing, and fraud detection.

3. Cluster Computing

- Cluster computing is a type of distributed computing that uses a group of interconnected computers to work together as a single system.
- Cluster computing is often used to solve problems that are too large or complex for a single computer to handle.

These are tightly coupled and homogenous



Cluster computing is used in a wide range of applications

- Financial Modelling: Financial institutions use cluster computing for risk assessment, portfolio optimization, and algorithmic trading. Complex financial models require significant computational resources, which clusters provide efficiently.
- Oil and Gas Exploration: The energy sector uses cluster computing for seismic data processing, reservoir simulation, and resource discovery. Cluster-based simulations help locate and extract oil and gas reserves more efficiently.
- Machine Learning and AI: Training large machine learning models and deep neural networks can be accelerated with cluster computing. Distributed training across multiple nodes can significantly reduce training times.

3. Grid Computing

- Grid computing is a type of distributed computing that uses a large number of geographically dispersed computers to work together as a single system
- The tasks they work on is of either high computing power and consist of large data sets.
- All communication between the computer systems in grid computing is done on the “data grid”.
- The goal of grid computing is to solve more high computational problems in less time and improve productivity.

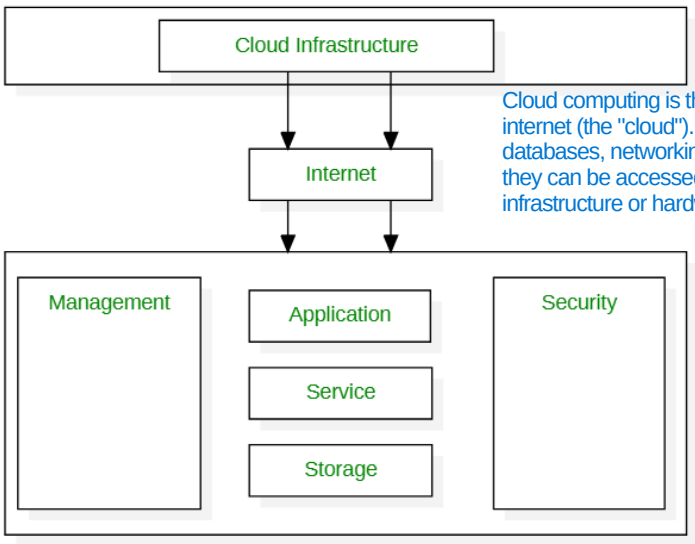
Unlike clusters, grids are loosely coupled and heterogeneous.

Grid computing is used in a wide range of applications

- Disaster Prediction and Response: Grid computing can help with earthquake simulations, tsunami modelling, and emergency response planning.
- Content Delivery and Content Distribution Networks (CDNs): CDNs use grid-like distributed architectures to deliver web content, videos, and other digital assets efficiently to users worldwide.
- Government and Defence: Grid computing is used for various defence applications, including cryptography, intelligence analysis, and simulations for military planning

4. Cloud Computing

- Cloud is defined as the usage of someone else’s server to host, process or store data.
- Cloud computing is defined as the type of computing where it is the delivery of on-demand computing services over the internet on a pay-as-you-go basis.
- It is widely distributed, network-based and used for storage.



Cloud computing is the delivery of computing services over the internet (the "cloud"). These services include servers, storage, databases, networking, software, analytics, and intelligence, and they can be accessed on-demand, without requiring local infrastructure or hardware.

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Comparison of various Computing Technologies

1. Difference between parallel and distributed

Parallel Computing	Distributed Computing
It is a type of computation in which various processes runs simultaneously	It is that type of computing in which the components are located on various networked systems that interact and coordinate their actions by passing messages to one another.
The processors communicate with one another via a bus	The computer systems connect with one another via a network.
Several processors execute various tasks simultaneously in parallel computing.	Several computers execute tasks simultaneously.
It occurs in a single computer system.	It involves various computers.
The system may have distributed or shared memory.	Each computer system in distributed computing has its own memory.
It helps to improve the system performance	It allows for scalability, resource sharing, and the efficient completion of computation tasks.

2. Difference between cloud, grid and cluster

Cloud	Grid	Cluster
Consolidation of resources	Segregation of resources	Aggregation of resources
Single system made up of many systems	Collection of systems that act together like a single system	Group of nodes that are connected to each other
Works with different hardware and OS	Works with different hardware and OS	All the connected systems should have same OS and hardware
Follows centralized architecture	Follows distributed architecture	Follows centralized architecture
Suffers from a single point of failure	All the nodes work independently, hence no single point of failure	Suffers from a single point of failure
The owners have less control over the systems on the cloud	The owners have full control and management over the grid's systems	Have no owners, and each node works independently
Job execution is self-managed	Scalability of execution allows for the transfer of a job's execution to an available processor	The scheduling of jobs affects execution. Jobs, therefore, wait until their designated runtime
Used in Dropbox, Gmail	Used in simulations, automations	Used in web servers, search engines

Cloud Computing

- Cloud computing can be defined as a new style of computing in which dynamically scalable and often virtualized resources are provided as a service over the Internet.

- Cloud computing is the delivery of computing services-including servers storge, database, networking, software analytics and intelligence - over the internet to offer faster innovation, flexible resources, and economies of scale.

History of cloud computing

- 1950s-1960s: Early conceptual roots of utility computing, envisioning shared remote computing resources.
- 1990s: Internet growth paves the way for remote resource sharing and web-based applications.
- Early 2000s: Term "cloud computing" gains popularity; Amazon AWS and Google offer cloud services.
- Late 2000s-Early 2010s: Rapid adoption of cloud computing for scalability, cost-efficiency; major providers like Microsoft Azure and IBM enter the market.
- Today, cloud computing is integral to IT, enabling scalability, cost reduction, and innovation across industries.

Features of cloud computing

1. Resources Pooling
   - It means that the Cloud provider pulled the computing resources to provide services to multiple customers with the help of a multi-tenant model.
   - There are different physical and virtual resources assigned and reassigned which depends on the demand of the customer.
   - The customer generally has no control or information over the location of the provided resources but is able to specify location at a higher level of abstraction.

2. On-Demand Self-Service
   - It is one of the important and valuable features of Cloud Computing as the user can continuously monitor the server uptime, capabilities, and allotted network storage.
   - With this feature, the user can also monitor the computing capabilities.

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3. Easy Maintenance
   - The servers are easily maintained and the downtime is very low and even in some cases, there is no downtime.
   - Cloud Computing comes up with an update every time by gradually making it better.
   - The updates are more compatible with the devices and perform faster than older ones along with the bugs which are fixed.

4. Large Network Access
   - The user can access the data of the cloud or upload the data to the cloud from anywhere just with the help of a device and an internet connection.
   - These capabilities are available all over the network and accessed with the help of internet.

5. Availability
   - The capabilities of the Cloud can be modified as per the use and can be extended a lot.

- It analyses the storage usage and allows the user to buy extra Cloud storage if needed for a very small amount.

6. Automatic System
   - Cloud computing automatically analyses the data needed and supports a metering capability at some level of services.
   - We can monitor, control, and report the usage.
   - It will provide transparency for the host as well as the customer.

7. Economical
   - It is the one-time investment as the company (host) has to buy the storage and a small part of it can be provided to the many companies which save the host from monthly or yearly costs.
   - Only the amount which is spent is on the basic maintenance and a few more expenses which are very less.

8. Security
   - Cloud Security, is one of the best features of cloud computing.
   - It creates a snapshot of the data stored so that the data may not get lost even if one of the servers gets damaged.
   - The data is stored within the storage devices, which cannot be hacked and utilized by any other person.
   - The storage service is quick and reliable.

9. Pay as you go
   - In cloud computing, the user has to pay only for the service or the space they have utilized.
   - There is no hidden or extra charge which is to be paid.
   - The service is economical and most of the time some space is allotted for free.

10. Measured Service
    - Cloud Computing resources used to monitor and the company uses it for recording.
    - This resource utilization is analysed by supporting charge-per-use capabilities.
    - This means that the resource usages which can be either virtual server instances that are running in the cloud are getting monitored measured and reported by the service provider.
    - The model pay as you go is variable based on actual consumption of the manufacturing organization

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Advantage and Disadvantages of Cloud Computing

Advantages of Cloud Computing

1) Back-up and restore data
   - Once the data is stored in the cloud, it is easier to get back-up and restore that data using the cloud.

. Environmentally Friendly
Energy Efficiency: Shared infrastructure reduces the carbon footprint compared to on-premises setups.

2) Improved collaboration

- Cloud applications improve collaboration by allowing groups of people to quickly and easily share information in the cloud via shared storage.

3) Excellent accessibility
- Cloud allows us to quickly and easily access store information anywhere, anytime in the whole world, using an internet connection. An internet cloud infrastructure increases organization productivity and efficiency by ensuring that our data is always accessible.

4) Low maintenance cost
- Cloud computing reduces both hardware and software maintenance costs for organizations.

5) Mobility
- Cloud computing allows us to easily access all cloud data via mobile.

6) Services in the pay-per-use model
- Cloud computing offers Application Programming Interfaces (APIs) to the users for access services on the cloud and pays the charges as per the usage of service.

7) Unlimited storage capacity
- Cloud offers us a huge amount of storing capacity for storing our important data such as documents, images, audio, video, etc. in one place.

8) Data security
- Data security is one of the biggest advantages of cloud computing. Cloud offers many advanced features related to security and ensures that data is securely stored and handled.

Disadvantages of Cloud Computing

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1) Internet Connectivity
   - As you know, in cloud computing, every data (image, audio, video, etc.) is stored on the cloud, and we access these data through the cloud by using the internet connection. If you do not have good internet connectivity, you cannot access these data. However, we have no any other way to access data from the cloud.

2) Vendor lock-in
   - Vendor lock-in is the biggest disadvantage of cloud computing. Organizations may face problems when transferring their services from one vendor to another. As different vendors provide different platforms, that can cause difficulty moving from one cloud to another.

3) Limited Control

- As we know, cloud infrastructure is completely owned, managed, and monitored by the service provider, so the cloud users have less control over the function and execution of services within a cloud infrastructure.

4) Security
   - Although cloud service providers implement the best security standards to store important information. But, before adopting cloud technology, you should be aware that you will be sending all your organization's sensitive information to a third party, i.e., a cloud computing service provider. While sending the data on the cloud, there may be a chance that your organization's information is hacked by Hackers

Applications of Cloud Computing:

1. Data Storage and Backup:
   - Cloud storage services like Amazon S3, Google Cloud Storage, and Microsoft Azure Blob Storage offer scalable and reliable data storage solutions. Organizations use these services to store and back up data securely.

2. Web Hosting and Development:
   - Cloud platforms provide infrastructure for web hosting, making it easier to deploy and manage websites and web applications. Services like AWS Elastic Beanstalk and Heroku simplify web application deployment.

3. Big Data and Analytics:
   - Cloud computing enables businesses to analyse vast amounts of data using services like Amazon Redshift, Google BigQuery, and Azure Data Lake Analytics. It allows for faster and cost-effective processing of big data workloads.

4. Machine Learning and AI:
   - Cloud providers offer machine learning platforms, such as Google Cloud AI, AWS SageMaker, and Azure Machine Learning, which allow developers and data scientists to build, train, and deploy machine learning models at scale.

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5. IoT (Internet of Things):
   - IoT devices generate massive amounts of data, which can be processed and analysed in the cloud. Cloud platforms provide the necessary infrastructure for managing IoT data and devices.

6. Content Delivery and Streaming:
   - Content delivery networks (CDNs) use cloud infrastructure to distribute content efficiently. Streaming services like Netflix and YouTube rely on CDNs to deliver content to users worldwide.

7. Gaming:
   - Cloud gaming services, such as Google Stadia and Microsoft's Xbox Cloud Gaming, leverage cloud resources to stream games to users' devices, reducing the need for powerful local hardware.

8. Collaboration and Productivity:

- Cloud-based collaboration tools like Microsoft 365 (formerly Office 365) and Google Workspace enable remote work, document sharing, and real-time collaboration among teams.

9. E-commerce:
   - Online retailers use cloud computing to host their e-commerce platforms, ensuring high availability, scalability, and security for their online stores.

10. Healthcare:
    - Cloud computing in healthcare facilitates the storage and sharing of electronic health records (EHRs), telemedicine services, medical research, and data analysis for improved patient care.
11. Finance and Fintech:
    - Financial institutions leverage cloud computing for risk management, algorithmic trading, fraud detection, and customer relationship management, while fintech startups use cloud resources to build and scale their platforms.

12. Government and Public Services:
    - Government agencies use cloud computing to improve citizen services, store sensitive data securely, and reduce IT infrastructure costs.

13. Education:
    - Cloud-based educational platforms provide remote learning opportunities, allowing students and educators to access resources and collaborate online.

14. Media and Entertainment:
    - Media companies use cloud services for content production, editing, and distribution. Streaming services, video rendering, and special effects rendering benefit from cloud scalability.

Trends in Cloud Computing

. Rise of Cloud Gaming
What it is: Delivering high-quality gaming experiences through cloud servers.

1. Edge computing:
   - Edge computing is a distributed computing paradigm that brings computing and data storage closer to the end-user. This can improve performance and reduce latency, especially for applications that require real-time processing of data.

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Cloud platforms offering AI/ML services for predictions, automation, and data analysis.

2. Artificial intelligence (AI) and machine learning (ML):
   - AI and ML are becoming increasingly important in cloud computing, as they can be used to automate tasks, improve decision-making, and personalize experiences.

3 Multi-cloud and hybrid cloud: Avoid vendor lock-in..Flexibility

   - Multi-cloud refers to the use of multiple cloud computing platforms from different providers, while hybrid cloud refers to the use of a combination of on-premises and cloud-based resources. These approaches can help organizations to improve flexibility, scalability, and security.

4. Serverless computing:

A cloud model where developers focus on code and business logic without managing servers or infrastructure.

   - Serverless computing is a cloud computing model in which the cloud provider manages the servers, storage, and networking resources needed to run an application. This can help organizations to reduce costs and complexity.

5. Data security and privacy:
   - As cloud computing becomes more widespread, the need for data security and privacy will become increasingly important. Cloud providers are investing heavily in security measures, but organizations also need to take steps to protect their data.

6. Blockchain:
   - Blockchain is a distributed ledger technology that can be used to record transactions securely and transparently. This makes it a promising technology for use in cloud computing, where it can be used to track data provenance, manage supply chains, and more.

7. Low-code and no-code development:
   - Low-code and no-code development platforms make it possible to create applications without writing code. This can help organizations to reduce development costs and time to market.

8. Web3:
   - Web3 is a new iteration of the internet that is built on blockchain technology. It has the potential to revolutionize cloud computing by making it more decentralized and secure.

Leading cloud platform services

1. Amazon Web Services (AWS):
   - AWS is the leading cloud platform provider, with a market share of over 34%. It offers a wide range of services, including compute, storage, networking, databases, analytics, machine learning, and artificial intelligence.

2. Microsoft Azure:
   - Azure is the second-largest cloud platform provider, with a market share of over 22%. It offers a similar range of services to AWS, but it also has a strong focus on hybrid cloud and enterprise applications.

3. Google Cloud Platform (GCP):
   - GCP is the third-largest cloud platform provider, with a market share of over 9.5%. It offers a wide range of services, including compute, storage, networking, machine learning, and artificial intelligence.

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4. Alibaba Cloud:
   - Alibaba Cloud is the fourth-largest cloud platform provider, with a market share of over 6%. It is the leading cloud platform in China, and it offers a wide range of services, including compute, storage, networking, big data, and artificial intelligence.

5. IBM Cloud:
   - IBM Cloud is the fifth-largest cloud platform provider, with a market share of over 4%. It offers a wide range of services, including compute, storage, networking, analytics, and machine learning.

Unit II : Cloud Architecture

Cloud service models:

Cloud services are a broad term that encompasses a wide range of services delivered over the internet. These services can be classified into three main categories:

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)

1. Infrastructure as a Service (IaaS):

High Control ,
High flexibility,
Used mostly by network architects

- Iaas is also known as Hardware as a Service (HaaS). It is one of the layers of the cloud computing platform.
- It allows customers to outsource their IT infrastructures such as servers, networking, processing, storage, virtual machines, and other resources.
- Customers access these resources on the Internet using a pay-as-per use model.
- EX:- Amazon Web Services (AWS), Microsoft Azure, Google Cloud, DigitalOcean, and Linode etc…
- IaaS provider provides the following services -
  - Compute: Computing as a Service includes virtual central processing units and virtual main memory for the Vms that is provisioned to the end-users.
  - Storage: IaaS provider provides back-end storage for storing files.
  - Network: Network as a Service (NaaS) provides networking components such as routers, switches, and bridges for the Vms.
  - Load balancers: It provides load balancing capability at the infrastructure layer.

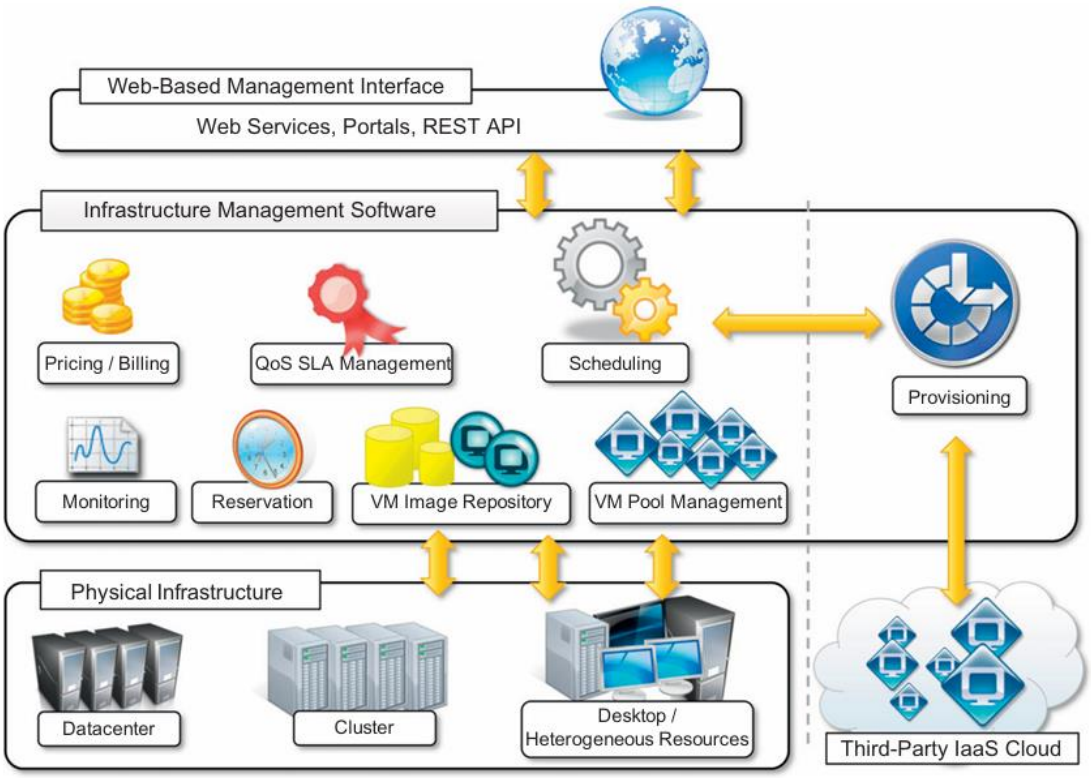


FIGURE 4.2

Infrastructure-as-a-Service reference implementation.

2. Platform as a Service (PaaS) :

- Provides a platform for developing, deploying, and managing applications.
- PaaS providers typically handle the underlying infrastructure, so PaaS customers can focus on developing their applications.
- Example: Google App Engine, Force.com, Joyent, Azure.

Development Focus: Developers can concentrate on coding and deploying applications.

Medium Control, moderately flexibility, used by developers

- PaaS providers provide the Programming languages, Application frameworks, Databases, and Other tools:
  1. Programming languages
     - PaaS providers provide various programming languages for the developers to develop the applications. Some popular programming languages provided by PaaS providers are Java, PHP, Ruby, Perl, and Go.
  2. Application frameworks
     - PaaS providers provide application frameworks to easily understand the application development. Some popular application frameworks provided by PaaS providers are Node.js, Drupal, Joomla, WordPress, Spring, Play, Rack, and Zend.
  3. Databases
     - PaaS providers provide various databases such as ClearDB, PostgreSQL, MongoDB, and Redis to communicate with the applications.
  4. Other tools
     - PaaS providers provide various other tools that are required to develop, test, and deploy the applications.

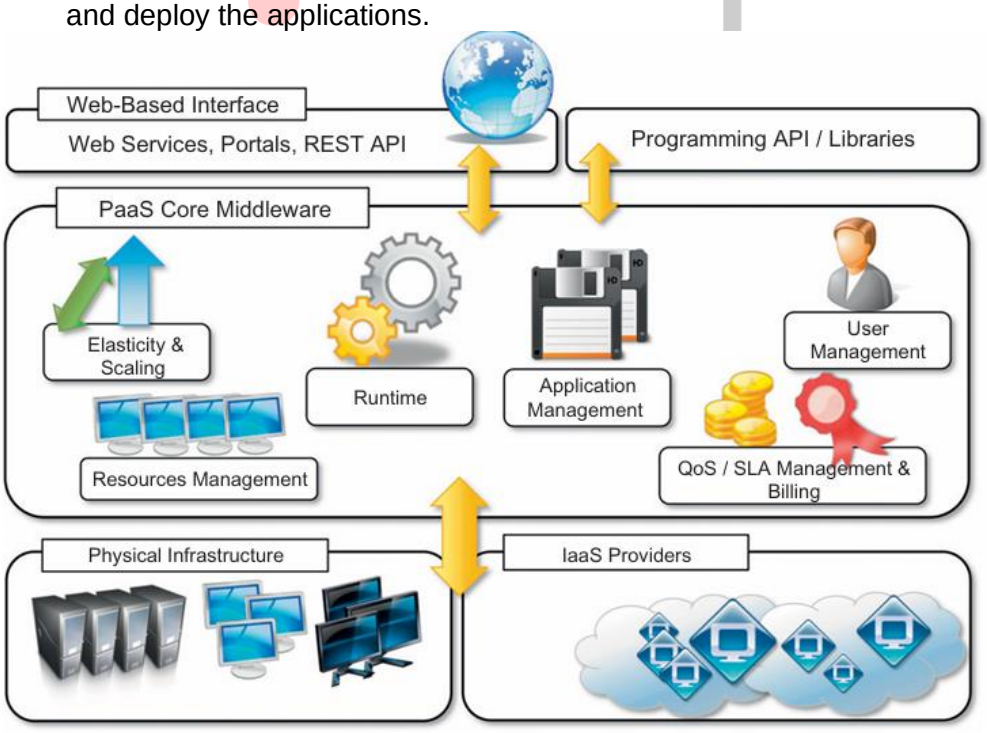


FIGURE 4.3

The Platform-as-a-Service reference model.

3. Software as a Service (SaaS):

less control, least flexible, used by end users

- SaaS is also known as "On-Demand Software". It is a software distribution model in which services are hosted by a cloud service provider.
- These services are available to end-users over the internet so, the end-users do not need to install any software on their devices to access these services.
- Ex:- Google Docs, Dropbox, etc…

- There are the following services provided by SaaS providers -

1. Business Services
   - SaaS Provider provides various business services to start-up the business.
   - The SaaS business services include ERP (Enterprise Resource Planning), CRM (Customer Relationship Management), billing, and sales.
2. Document Management
   - SaaS document management is a software application offered by a third party (SaaS providers) to create, manage, and track electronic documents.
   - Example: Slack, Samepage, Box, and Zoho Forms.
3. Social Networks
   - As we all know, social networking sites are used by the general public, so social networking service providers use SaaS for their convenience and handle the general public's information.
4. Mail Services
   - To handle the unpredictable number of users and load on e-mail services, many e-mail providers offering their services using SaaS.

Comparison of different Service Models;

definition,
services provided(servers - programming languges - mail services)
used by
control
flexibility
skill requirement
Examples

IaaS	Paas	SaaS
It provides a virtual data center to store information and create platforms for app development, testing, and deployment.	It provides virtual platforms and tools to create, test, and deploy apps.	It provides web software and apps to complete business tasks.
It provides access to resources such as virtual machines, virtual storage, etc.	It provides runtime environments and deployment tools for applications.	It provides software as a service to the end-users.
It is used by network architects.	It is used by developers.	It is used by end users.
IaaS provides only Infrastructure.	PaaS provides Infrastructure+Platform.	SaaS provides Infrastructure+Platform +Software

Skill Requirement High – Requires expertise in infrastructure management. Moderate – Knowledge of application development is needed. Low – Minimal technical knowledge is sufficient.

Cloud Deployment Models:

- Cloud Deployment Model functions as a virtual computing environment with a deployment architecture that varies depending on the amount of data you want to store and who has access to the infrastructure.
- These Models can be classified into four main categories:
  1. Public cloud
  2. Private cloud
  3. Hybrid cloud
  4. Community cloud

Cloud deployment models define how the cloud services are made available to users

The public cloud is a cloud computing model in which services such as servers, storage, and applications are offered over the internet by third-party cloud providers. These services are accessible to anyone who wants to use them, and they operate on a shared infrastructure.

1. **Public cloud:**
   - The public cloud is available to the general public, and resources are shared between all users.
   - Public clouds are the most common type of cloud deployment model. In a public cloud, the infrastructure is owned and managed by a cloud service provider (CSP) such as Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP).
   - Customers can access public cloud resources over the internet on a pay-as-you-go basis. Example: Amazon Web services

Advantages of public cloud:

- Low cost: Public clouds are typically very cost-effective, as customers only pay for the resources they use.
- Scalability and flexibility: Public clouds are highly scalable and flexible, allowing customers to quickly add or remove resources as needed.
- Ease of use: Public clouds are generally easy to use, with self-service portals and tools that make it easy to manage cloud resources.
- No maintenance: The maintenance work is done by the service provider (not users).

Disadvantages of public cloud:

- Security and privacy: Some customers may be concerned about the security and privacy of their data in a public cloud. However, most CSPs offer a variety of security features and controls to help customers protect their data.
- Lack of control: Customers have less control over the underlying infrastructure in a public cloud.
- Vendor lock-in: It can be difficult and expensive to switch cloud providers once you have migrated your applications and data to a public cloud.

A private cloud is a cloud infrastructure dedicated to a single organization. It can be hosted on-premises or by a third-party provider but is used exclusively by one entity.

2. **Private cloud:**
   - The private cloud deployment model is the exact opposite of the public cloud deployment model.
   - The private cloud deployment model is a dedicated environment for one user or customer. You don't share the hardware with any other users
   - Private clouds can be deployed on-premises or hosted by a CSP.

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offering more control, security, and customization. Eg: Openstack,VmWare

Advantages of private cloud:

- Security and privacy: Private clouds offer a higher level of security and privacy than public clouds, as the infrastructure is dedicated to a single customer or organization.
- Control: Customers have more control over the underlying infrastructure in a private cloud.
- Customization: Private clouds can be customized to meet the specific needs of the customer or organization.

Disadvantages of private cloud:

- Higher cost: Private clouds are typically more expensive than public clouds.
- Less scalability: Private clouds are not as scalable as public clouds, as the customer or organization is responsible for managing the infrastructure.
- Complexity: Private clouds can be more complex to manage than public clouds.

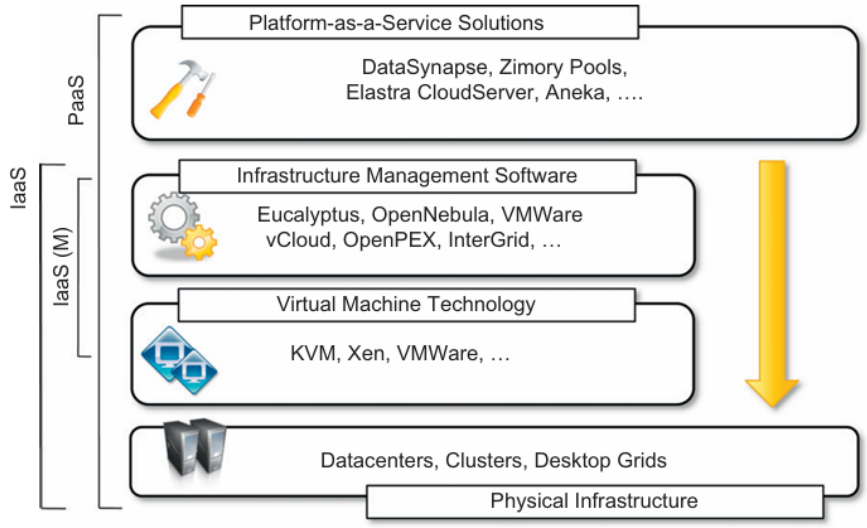


FIGURE 4.4

Private clouds hardware and software stack.

3. **Hybrid cloud:** A hybrid cloud combines public and private clouds, allowing organizations to share data and applications between the two environments.
   - A hybrid cloud deployment model combines public and private clouds.
   - Creating a hybrid cloud computing model means that a company uses the public cloud but owns on-premises systems and provides a connection between the two. Eg: Microsoft Azure stack
   - They work as one system, which is a beneficial model for a smooth transition into the public cloud over an extended period.

   Advantages of hybrid cloud:
   - Flexibility: Hybrid clouds offer the flexibility to choose the best cloud deployment model for each workload.
   - Scalability: Hybrid clouds can be scaled to meet the changing needs of the customer or organization.
   - Cost-effectiveness: Hybrid clouds can be more cost-effective than private clouds, as the customer or organization can use public cloud resources for non-critical workloads.

   Disadvantages of hybrid cloud:
   - Complexity: Hybrid clouds can be more complex to manage than public or private clouds.
   - Security and privacy: Customers need to carefully manage the security and privacy of their data in a hybrid cloud environment.

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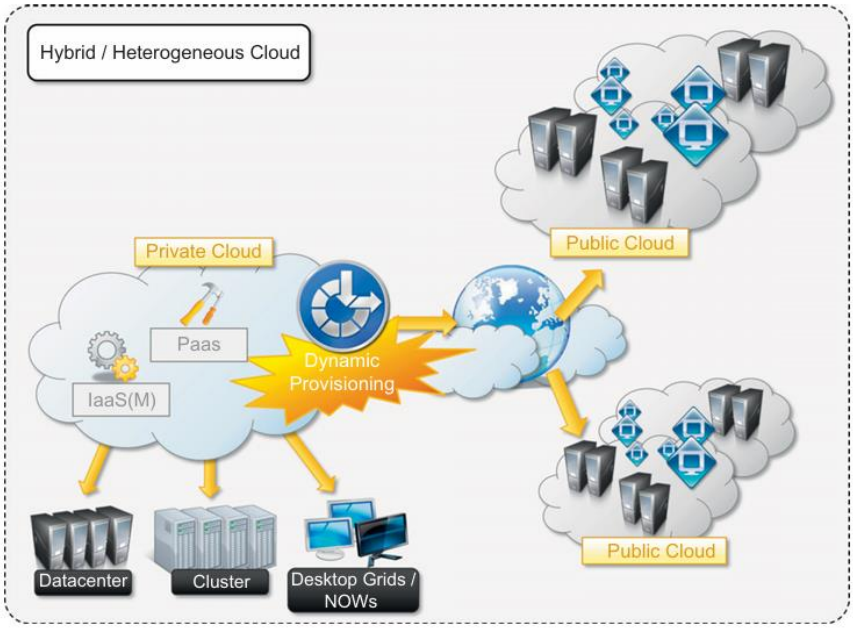


FIGURE 4.5

Hybrid/heterogeneous cloud overview.

4. **Community cloud:** A community cloud is a shared environment designed for organizations with common interests, such as industry regulations or security requirements. It is owned and managed by one or more of the participating organizations or a third party.
   - The community cloud operates in a way that is similar to the public cloud.
   - There's just one difference - it allows access to only a specific set of users who share common objectives and use cases.
   - Community clouds are shared by a group of organizations with similar needs. Community clouds can be deployed on-premises or hosted by a CSP.

Healthcare organizations collaborating on patient data systems

Advantages of community cloud:

- Cost-effectiveness: Community clouds can be more cost-effective than private clouds, as the cost of the infrastructure is shared among the participating organizations.
- Security and privacy: Community clouds offer a higher level of security and privacy than public clouds, as the infrastructure is shared by a group of trusted organizations.
- Collaboration: Community clouds can facilitate collaboration between the participating organizations.

Disadvantages of community cloud:

- Limited scalability: Community clouds are not as scalable as public clouds, as the customer or organization is responsible for managing the infrastructure.
- Complexity: Community clouds can be more complex to manage than public clouds.
- Limited availability: Community clouds are less common than public and private clouds.

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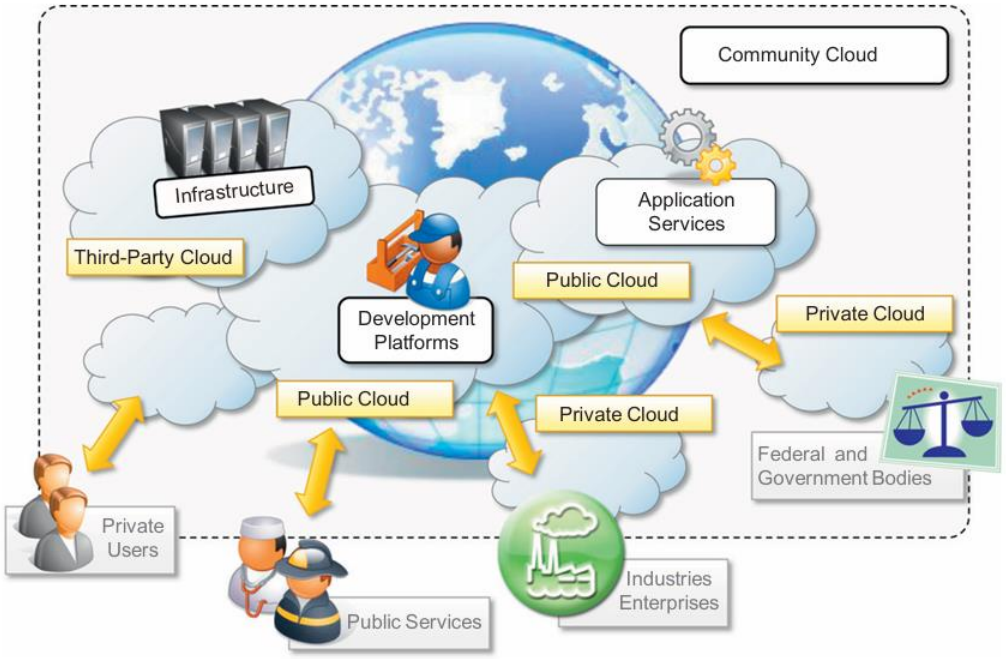


FIGURE 4.6

A community cloud.

Cloud Computing Architecture

- Cloud computing architecture is the conceptual design and fundamental operational structure of a cloud computing system.
- It is a high-level abstraction that describes the major components of the cloud and how they interact with each other.

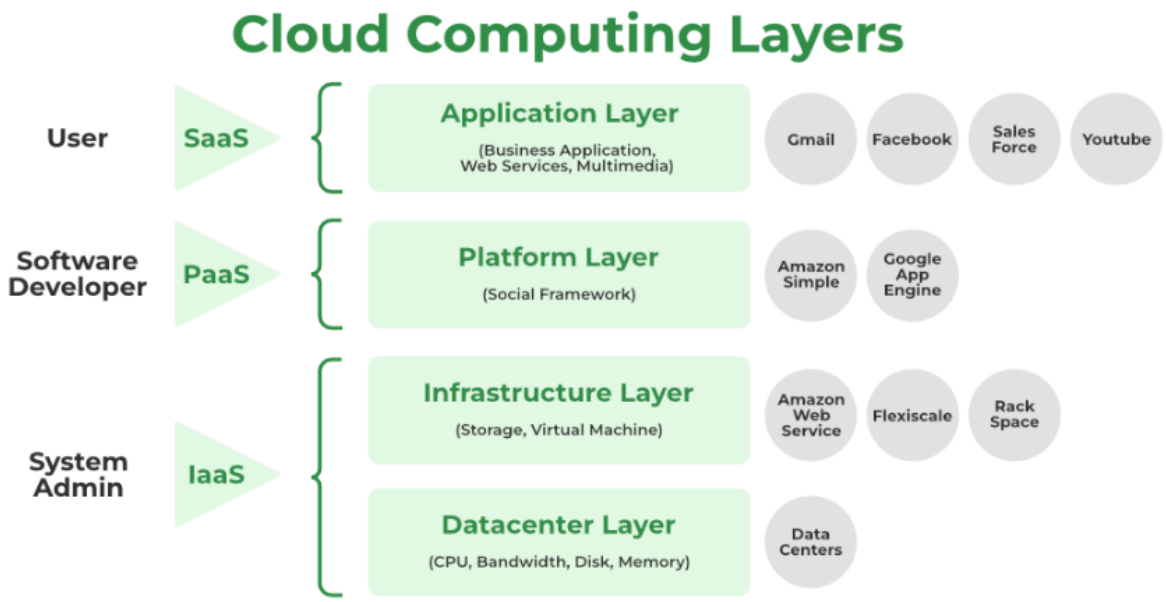
Benefits of cloud computing architecture:

- Scalability: Cloud computing architecture is designed to be scalable, so businesses can easily add or remove resources as needed.
- Agility: Cloud computing architecture allows businesses to deploy and manage applications more quickly and easily than traditional on-premises IT systems.
- Cost-effectiveness: Cloud computing architecture can help businesses reduce their IT costs by eliminating the need to purchase and maintain their own hardware and software infrastructure.
- Reliability: Cloud computing architecture is designed to be reliable, with high availability and disaster recovery capabilities.
- Security: Cloud computing architecture can provide a high level of security for businesses and their data.

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Layered Architecture of Cloud:

- The layered architecture of cloud computing is a conceptual design that describes the major components of the cloud and how they interact with each other.
- It is typically divided into three layers:
  - Application layer.
  - Platform layer.
  - Infrastructure layer.
  - Datacentre layer.



1. **Application layer:**
   - The application layer is the top layer of the cloud architecture. It includes the cloud applications that are used by end users.
   - Users can access these applications according to their needs. Applications are divided into Execution layers and Application layers.
   - In order for an application to transfer data, the application layer determines whether communication partners are available. Whether enough cloud resources are accessible for the required communication is decided at the application layer. Applications must cooperate in order to communicate, and an application layer is in charge of this.
   - The application layer, in particular, is responsible for processing IP traffic handling protocols like Telnet and FTP.
   - Other examples of application layer services include web browsers, SNMP protocols, HTTP protocols, or HTTPS, which is HTTP's successor protocol.

2. **Platform layer:**
   - The platform layer sits on top of the infrastructure layer and provides a platform for developing, deploying, and managing cloud applications. It includes services such as operating systems, databases, middleware, and development tools.
   - The operating system and application software make up this layer.
   - Users should be able to rely on the platform to provide them with Scalability, Dependability, and Security Protection which gives users a space to create their apps, test operational processes, and keep track of execution outcomes and performance.
   - The objective of this layer is to deploy applications directly on virtual machines.
   - The platform layer's goal is to lessen the difficulty of deploying programmers directly into VM containers.
   - By way of illustration, Google App Engine functions at the platform layer to provide API support for implementing storage, databases, and business logic of ordinary web apps.

3. **Infrastructure layer:**
   - The infrastructure layer is the bottom layer of the cloud architecture. It consists of the physical hardware and software resources that support the cloud, such as servers, storage, networking, and virtualization software.

- It is a layer of virtualization where physical resources are divided into a collection of virtual resources using virtualization technologies like Xen, KVM, and VMware.
- This layer serves as the Central Hub of the Cloud Environment, where resources are constantly added utilizing a variety of virtualization techniques.
- The infrastructure layer sometimes referred to as the virtualization layer, partitions the physical resources using virtualization technologies like Xen, KVM, Hyper-V, and VMware to create a pool of compute and storage resources.

4. **Datacenter layer:**
   - In a cloud environment, this layer is responsible for Managing Physical Resources such as servers, switches, routers, power supplies, and cooling systems.
   - Providing end users with services requires all resources to be available and managed in data center
   - Physical servers connect through high-speed devices such as routers and switches to the data center.

Virtualization in cloud computing is a technology that enables the creation of virtual versions of physical resources, such as servers, storage devices, networks, or even entire operating systems. By using virtualization, a single physical machine can host multiple virtual machines (VMs), each running its own operating system and applications, independently from each other

Virtualization

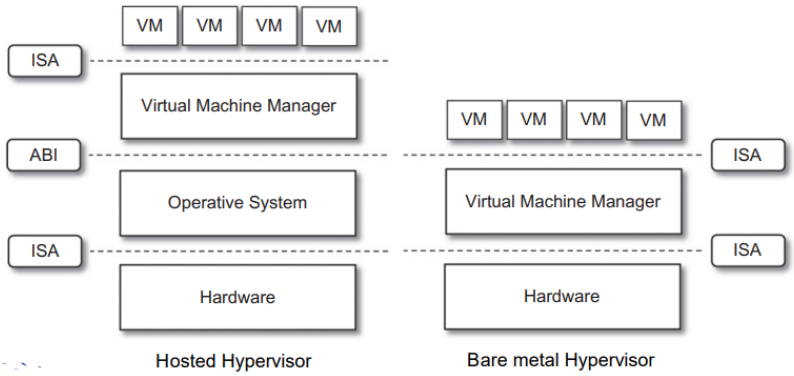
- Virtualization is a process that allows a computer to share its hardware resources with multiple digitally separated environments. Each virtualized environment runs within its allocated resources, such as memory, processing power, and storage.

Important terms in virtualization.

1. Virtual machine
   - A virtual machine is a software-defined computer that runs on a physical computer with a separate operating system and computing resources.
   - Multiple virtual machines can run on a single physical machine.
   - Virtual machines are abstracted from the computer hardware by a hypervisor.

2. Hypervisor
   - A hypervisor is a software that you can use to run multiple virtual machines on a single physical machine.
   - Every virtual machine has its own operating system and applications.
   - The hypervisor allocates the underlying physical computing resources such as CPU and memory to individual virtual machines as required.
   - There are two major types of hypervisor: Type I and Type II
     - Type1:- The hypervisor runs directly on the underlying host system. It is also known as a “Native Hypervisor” or “Bare metal hypervisor”. It does not require any base server operating system. It has direct access to hardware resources.
     - Type2:- A Host operating system runs on the underlying host system. It is also known as ‘Hosted Hypervisor”. Such kind of hypervisors doesn’t run directly over the underlying hardware rather they run as an application in a Host system(physical machine)

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3. Host Machine

- The machine on which the virtual machine is create is known as Host Machine.

4. Guest Machine

- The virtual machines which are created on Host Machine is called Guest Machine.

Some of virtualization in cloud computing examples are as follows:

- EC2 service from Amazon Web Service.
- Compute engine from Google Cloud.
- Azure Virtual Machines from Microsoft Azure.

Characteristics\Features of Virtualization

Platform Independence
Definition: VMs can run on different types of hardware or OS platforms

Scalability

1. Managed Resources:
   - The VMs or any resources provisioned from the cloud are completely managed by the cloud providers, meaning apart from specifying the requirements, users don't have to maintain or worry about the underlying hardware and configuration of their resources.
   - For example, the cloud providers handle all the work such as hardware supplies, backups, monitoring, etc.

2. Resource Allotment:
   - The resource allotment is made simple with virtualization technology, the process is as simple as clicking a few buttons.
   - Users can get their whole infrastructure ready within hours, and they can customize it later without any hassle.

3. Resource Isolation
   - Resource isolation is one of the important characteristics of virtualization in the cloud. This enables applications to run in an environment that is completely dedicated to them. This also helps in preventing data breaches and makes sure to enable efficient resource utilization.

4. Load balancing
   - The load balancer as the name suggest handles the request load of a server. The virtualization platforms make sure that the incoming requests are distributed to the appropriate servers. This allows the servers to serve requests sooner instead of being overloaded by the huge number of requests.

5. Portability

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- The virtualization resources are portable, meaning they can be copied and moved from one system to another, and the same functionality can be expected. This allows the users to create and reuse the configuration instead of repeating it.

Types of Virtualizations:

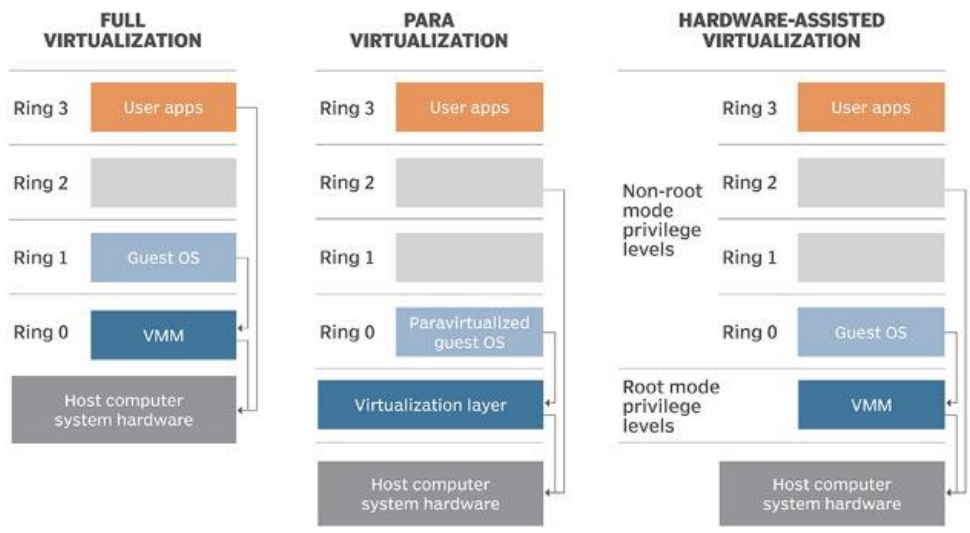
1. Hardware virtualization
2. Server virtualization
3. Application virtualization
4. Storage virtualization
5. Operating System virtualization

1. Hardware virtualization:

- Hardware-level virtualization is a virtualization technique that provides an abstract execution environment in terms of computer hardware on top of which a guest operating system can be run. In this model, the guest is represented by the operating system, the host by the physical computer hardware, the virtual machine by its emulation, and the virtual machine manager by the hypervisor.

Hypervisors

- A fundamental element of hardware virtualization is the hypervisor, or virtual machine manager (VMM). It recreates a hardware environment in which guest operating systems are installed. There are two major types of hypervisor: Type I and Type II.



a. Full virtualization:

- Full virtualization refers to the ability to run a program, most likely an operating system, directly on top of a virtual machine and without any modification, as though it were run on the raw hardware.
- To make this possible, virtual machine manager are required to provide a complete emulation of the entire underlying hardware.
- The principal advantage of full virtualization is complete isolation, which leads to enhanced security, ease of emulation of different architectures, and coexistence of different systems on the same platform.

b. Paravirtualization:

- Runs a modified and recompiled version of the guest OS in a VM. This modification enables the VM to differ somewhat from the hardware.
- The hardware isn't necessarily simulated in paravirtualization but uses an application program interface (API) that can modify guest OS.
- To modify the guest OS, the source code for the guest OS must be accessible to replace portions of code with customizable instructions, such as calls to VMM APIs. The OS is then recompiled to use the new modifications.

c. Hardware-assisted virtualization:

- This term refers to a scenario in which the hardware provides architectural support for building a virtual machine manager able to run a guest operating system in complete isolation.
- This technique was originally introduced in the IBM System/370. At present, examples of hardware-assisted virtualization are the extensions to the x86-64 bit architecture introduced with Intel VT (formerly known as Vanderpool) and AMD V (formerly known as Pacifica).

2. Server virtualization:

- Server virtualization helps organizations to partition server resources in a way that ensures full utilization of the resources. One of its primary goals is that it helps in breaking up huge physical servers into several instances of virtual servers. This makes it possible for each server to be masked and run as a standalone server.

Advantages if Server virtualization

- Resource Efficiency: Server virtualization enables better utilization of hardware resources. By running multiple virtual servers on a single physical server, you can maximize CPU, memory, storage, and network capacity, reducing hardware and operational costs.
- Cost Savings: Fewer physical servers are needed, leading to reduced capital expenses for purchasing and maintaining hardware. Additionally, energy consumption and cooling costs are lower, contributing to significant cost savings over time.
- Isolation and Security: Virtual servers can be isolated from each other, enhancing security. If one virtual server experiences a security breach or a crash, it doesn't affect others. This isolation improves overall system stability and security.

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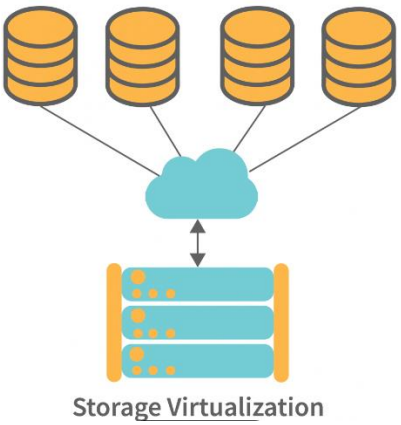
3. Application virtualization:

- Application virtualization pulls out the functions of applications to run on operating systems other than the operating systems for which they were designed.
- For example, users can run a Microsoft Windows application on a Linux machine without changing the machine configuration.
- To achieve application virtualization, follow these practices:
  - Application streaming – Users stream the application from a remote server, so it runs only on the end user's device when needed.
  - Server-based application virtualization – Users can access the remote application from their browser or client interface without installing it.
  - Local application virtualization – The application code is shipped with its own environment to run on all operating systems without changes.

4. Storage virtualization:

- Storage virtualization combines the functions of physical storage devices such as network attached storage (NAS) and storage area network (SAN).

- You can pool the storage hardware in your data centre , even if it is from different vendors or of different types.
- Storage virtualization uses all your physical data storage and creates a large unit of virtual storage that you can assign and control by using management software.
- IT administrators can streamline storage activities, such as archiving, backup, and recovery, because they can combine multiple network storage devices virtually into a single storage device.
- There are different techniques for storage virtualization, one of the most popular being network-based virtualization by means of storage area networks (SANs).
- SANs use a network-accessible device through a large bandwidth connection to provide storage facilities.



5. Operating System virtualization:
   - Operating system-level virtualization offers the opportunity to create different and separated execution environments for applications that are managed concurrently.
   - Differently from hardware virtualization, there is no virtual machine manager or hypervisor, and the virtualization is done within a single operating system, where the OS kernel allows for multiple isolated user space instances.
   - The kernel is also responsible for sharing the system resources among instances and for limiting the impact of instances on each other.
   - Examples of operating system-level virtualizations are FreeBSD Jails, IBM Logical Partition (LPAR), SolarisZones and Containers, Parallels Virtuozzo Containers, OpenVZ, iCore Virtual Accounts, Free Virtual Private Server (FreeVPS), and others.
   - The services offered by these technologies differ, and most of them are available on Unix-based systems. Some of them, such as Solaris and OpenVZ, allow for different versions of the same operating system to operate concurrently.

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Pros & Cons of Virtualization in Cloud Computing

Advantages(Pros) of Virtualization

1. Reduced Cost
   - Virtualization in the cloud provides an easy-to-use platform, which enables the users to provision resources with few clicks, and pay for only what they use.
   - Due to this, the user doesn't have to set up their own physical server and handle the maintenance, which would be expensive.

2. Increased Uptime
   - Virtualization in the cloud enables users to set up virtual resources in many locations around the world as a backup.

- This increases the uptime and availability of the resources and the user doesn't have to worry if one or more of their servers goes down, since there are backup resources available.

3. Increased Security
   - Cloud providers take extra measures to secure every single resource they provide, and these security measures are implemented layer by layer, starting from the hardware level to the software level.
   - The security measures include Firewall to defend against cyber and virus threats, End-to-End encryption, Data Backups, and more. The cloud providers let users define some of their security policies as well, such as adding a member, whitelisting IP addresses, and more.

4. Flexible Provisioning
   - The resourcing provisioning in the cloud is as simple as clicking a few buttons. Users specify the type of resource and the capacity of resources they need, and those requested sets of resources will be provided within a few minutes.

5. Easy setup
   - Virtualization in cloud makes sure that the platform is easy to use for the users who are setting up their infrastructure.

6. Utilization of Hardware Efficiently
   - With the help of virtualization, hardware is efficiently used by the user as well as the cloud service provider. This results in less cost for the user and the service provider.

7. Efficient and Easy Disaster Recovery
   - With the help of virtualization, data recovery, backup, and duplication become very easy.

Disadvantages (Cons) of Virtualization

1. Data Privacy Concerns
   - Along with all the benefits of the cloud and virtualization, one of the important disadvantages is privacy, because even though virtualization in the cloud would help users to create any number of virtual resources, the data and all the activities are stored and managed by a third party.
   - This wouldn't be a concern for all the application, but for some it does. Cloud providers often come up with agreements, encryption, and many other ways to overcome this.

2. Learning Curve
   - Mastering virtualization technology in cloud has a bit difficult learning curve and can take some time and experience, since along with creating resources one must handle the network configuration, defining policies, whitelisting IPs, deploying applications, etc.

3. Increased Risk of Over Billing
   - Overbilling is a common concern in the cloud, this usually depends on how the user provisions the resources and uses them. Some users may forget to terminate their resources or increases the resource capacity which increases the billing amount.

4. Possibilities of Vendor Locking

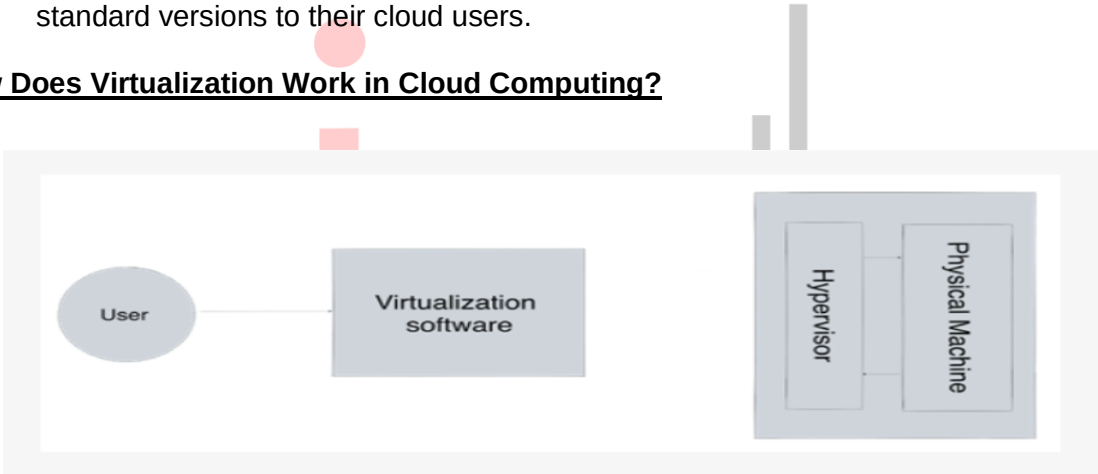
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- Vendor locking is one of the ways where cloud providers restrict users from moving out of their platform. The whole platform is designed in such a way that all the services depend on one another, and due to this it is hard for the user to move out of one platform to another.
- This is one of the biggest issues when there's a downtime, meaning for some unexpected reason if the cloud service goes down, then all the services that are being used by the users goes down as well.

Virtualization and Cloud Computing

- Virtualization is the creation of a virtual version of something, such as a server, a desktop, a storage device, an operating system or network resources.
- It allows sharing a single physical instance of a resource or an application among multiple customers and organizations by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.
- Virtualization plays a very important role in cloud computing technology.
- Normally in cloud computing, users share the data present in the clouds like applications etc, but actually with the help of virtualization users share the infrastructure.
- The main usage of Virtualization Technology is to provide the applications with the standard versions to their cloud users.

How Does Virtualization Work in Cloud Computing?



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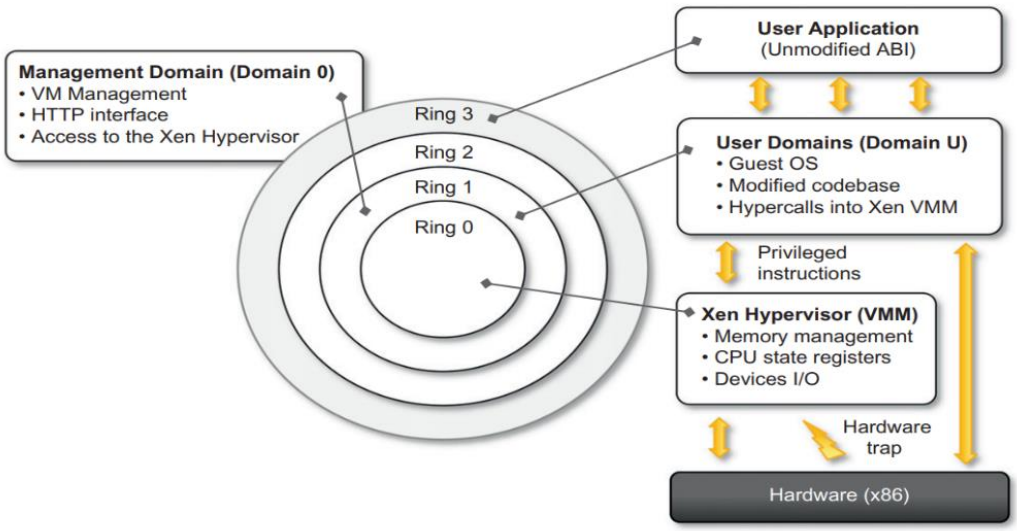
- Virtualization in the cloud is achieved by combining multiple layers of virtualization, these layers include hardware/physical, network, storage, and application virtualization. These layers can be customized and controlled by the cloud software by the users.
- The interaction with virtualization starts with the users sending a request for a virtual resource via the cloud software, which transfers the request to the hypervisor. The hypervisor then checks for the available resources and allocates the requested amount of resources to the user.
- Due to this simplified interaction, the users can set up their entire infrastructure in less time and with less maintenance instead of having to deal with setting up the entire physical infrastructure and maintaining it themselves.

Technology Examples

Xen- Paravirtualization

- Xen is an open-Source Hypervisor based on Paravirtualization.
- Xen is a type1 hypervisor, i.e., it runs on hardware itself.

- Xen-based technology is used for either desktop virtualization or server virtualization.
- Recently it has also been used to provide cloud computing solutions by means of Xen cloud platform (XCP).
- Xen is the most popular implementation of paravirtualization, which in contrast with full virtualization, allows high-performance execution of guest operating system.
- This is done by modifying portions of the guest operating system run by Xen with reference to the execution of privileged instructions.
- A Xen-based system is managed by the Xen hypervisor, which runs in the highest privileged mode and controls the access of guest operating system to the underlying hardware. Guest operating systems are executed within domains, which represent virtual machine instances.
- specific control software, which has privileged access to the host and controls all the other guest operating systems, is executed in a special domain called Domain 0.
- This is the first one that is loaded once the virtual machine manager has completely booted, and it hosts a HyperText Transfer Protocol (HTTP) server that serves requests for virtual machine creation, configuration, and termination.
- Below figure describes the architecture of Xen and its mapping onto a classic x86 privilege model.



- Many of the x86 implementations support four different security levels, called rings, ring 0, ring 1, ring 2, ring 3.
- Ring 0 represent the level with the high privileges and Ring 3 the level with the lowest ones.
- Almost all the frequently used operating system, except for operating system /2, uses only two levels. i.e., Ring 0 for the kernel and ring 3 for user application and non-privilege operating system code.
- Because of the structure of the x86. Some instruction allow code executing in ring 3 to jump into ring 0 (Kernel mode).
- Such operation is performed at the hardware level and therefore within the virtualized environment will result in a trap or silent fault, thus preventing the normal operations of the guest operating system.
- This condition is generally triggered by a subset of system calls.

- To avoid this situation, operating system needs to be changed in their implementation, and the sensitive system calls need to be reimplemented with hyper calls.
- With the use of hyper calls, the xen hypervisor is able to catch the execution of all the sensitive instruction, manage them, and return the control to the guest operating system.

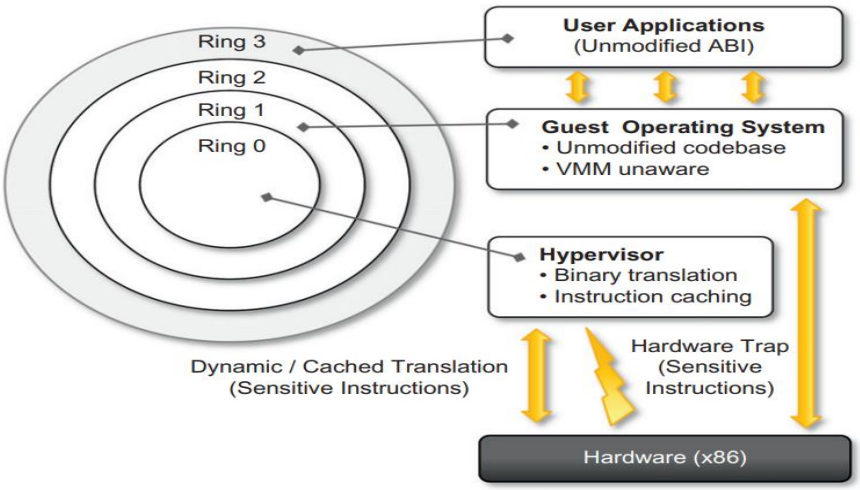
VMware- full virtualization

- VMware's technology is based on the concept of full virtualization, where the underlying hardware is replicated and made available to the guest operating system, which runs unaware of such abstraction layers and does not need to be modified.
- VMware implements full virtualization either in the desktop environment, by means of Type II hypervisors, or in the server environment, by means of Type I hypervisors.
- In both cases, full virtualization is made possible by means of direct execution (for non sensitive instructions) and binary translation (for sensitive instructions), thus allowing the virtualization of architecture such as x86.

Full virtualization and binary translation

- VMware is well known for the capability to virtualize x86 architectures, which runs unmodified on top of their hypervisors.
- x86 architecture design does not satisfy the first theorem of virtualization, since the set of sensitive instructions is not a subset of the privileged instructions.
- This causes a different behaviour when such instructions are not executed in Ring 0, which is the normal case in a virtualization scenario where the guest OS is run in Ring 1.
- Generally, a trap is generated and the way it is managed differentiates the solutions in which virtualization is implemented for x86 hardware.
- In the case of dynamic binary translation, the trap triggers the translation of the offending instructions into an equivalent set of instructions that achieves the same goal without generating exceptions.
- Moreover, to improve performance, the equivalent set of instruction is cached so that translation is no longer necessary for further occurrences of the same instructions. Below diagram gives an idea of the process.
- The major advantage is that guests can run unmodified in a virtualized environment, On the other hand, translating instructions at runtime introduces an additional overhead that is not present in other approaches.
- Disadvantage exists, binary translation is applied to only a subset of the instruction set, whereas the others are managed through direct execution on the underlying hardware.

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A full Virtualization reference model

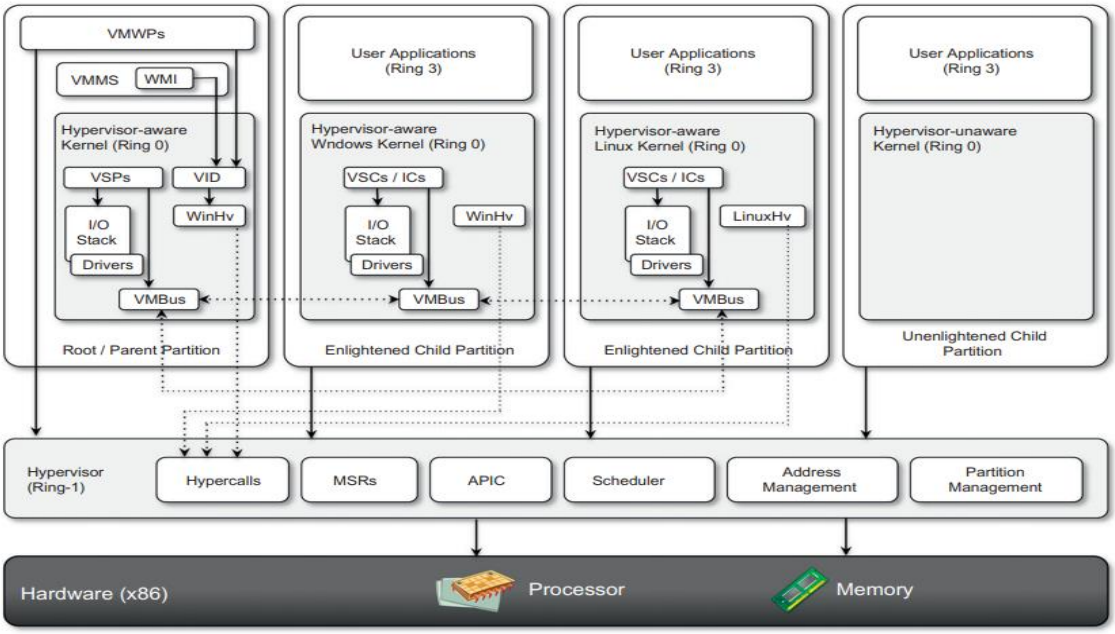
Microsoft Hyper-V

- Hyper-V is an infrastructure virtualization solution developed by Microsoft for server virtualization.

Architecture

- Hyper-V supports multiple and concurrent execution of guest operating systems by means of partitions.
- A partition is a completely isolated environment in which an operating system is installed and run.
- Despite its straightforward installation as a component of the host operating system, Hyper-V takes control of the hardware, and the host operating system becomes a virtual machine instance with special privileges, called the parent partition.
- The parent partition (also called the root partition) is the only one that has direct access to the hardware.
- It runs the virtualization stack, hosts all the drivers required to configure guest operating systems, and creates child partitions through the hypervisor.
- Child partitions are used to host guest operating systems and do not have access to the underlying hardware, but their interaction with it is controlled by either the parent partition or the hypervisor itself.
- Below diagram, provides an overview of the architecture of Hyper-V.

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Parent partition

- The parent partition executes the host operating system and implements the virtualization stack that complements the activity of the hypervisor in running guest operating systems.
- This partition is the only one that directly accesses device drivers and mediates the access to them by child partitions by hosting the VSPs.
- The parent partition is also the one that manages the creation, execution, and destruction of child partitions.
- It does so by means of the Virtualization Infrastructure Driver (VID), which controls access to the hypervisor and allows the management of virtual processors and memory.
- For each child partition created, a Virtual Machine Worker Process (VMWP) is instantiated in the parent partition, which manages the child partitions by interacting with the hypervisor through the VID.
- Virtual Machine Management services are also accessible remotely through a WMI9 provider that allows remote hosts to access the VID.

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Child partitions

- Child partitions are used to execute guest operating systems. These are isolated environments that allow secure and controlled execution of guests. Two types of child partition exist, they differ on whether the guest operating system is supported by Hyper-V or not.

Hypervisor

- The hypervisor is the component that directly manages the underlying hardware (processors and memory).
- It is logically defined by the following components:
    1. Hypercalls interface.
        - This is the entry point for all the partitions for the execution of sensitive instructions.

- This interface is used by drivers in the partitioned operating system to contact the hypervisor using the standard Windows calling convention.
- The parent partition also uses this interface to create child partitions.

2. Memory service routines (MSRs).
   - These are the set of functionalities that control the memory and its access from partitions.
3. Advanced programmable interrupt controller (APIC).
   - This component represents the interrupt controller, which manages the signals coming from the underlying hardware when some event occurs (timer expired, I/O ready, exceptions and traps).
4. Scheduler.
   - This component schedules the virtual processors to run on available physical processors.
   - The scheduling is controlled by policies that are set by the parent partition.
5. Address manager.
   - This component is used to manage the virtual network addresses that are allocated to each guest operating system.
6. Partition manager.
   - This component is in charge of performing partition creation, finalization, destruction, enumeration, and configurations.
   - Its services are available through the hypercalls interface API previously discussed.

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Aneka is a cloud application development platform created by Manjrasoft. It provides a framework that helps developers build, run, and manage cloud-based applications by leveraging the computing resources of local machines, private clouds, and public cloud platforms.

Think of Aneka as a toolbox for developers to create applications that can run in a cloud environment. Instead of worrying about how to connect to cloud servers or how to use multiple computers efficiently, developers use Aneka to handle those technical tasks for them

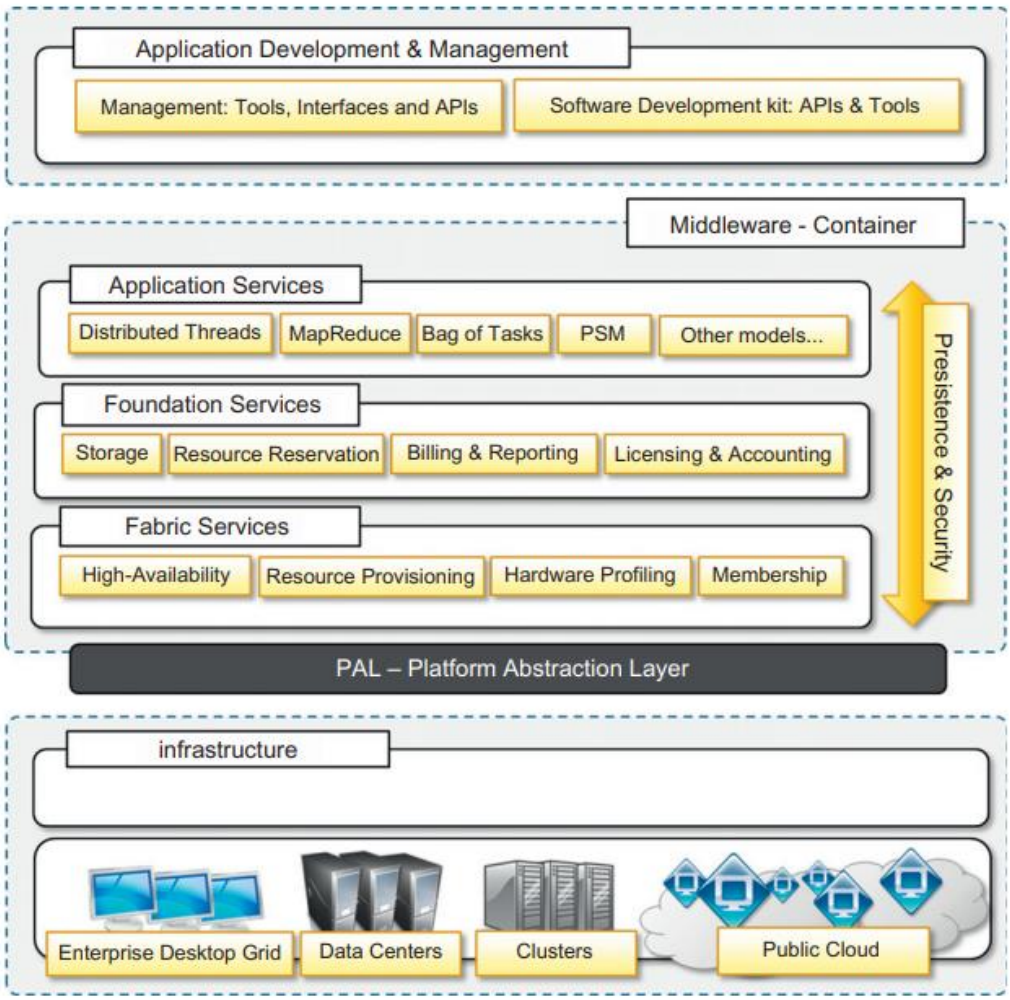
Unit III: Cloud Application programming and the Aneka Platform

Aneka cloud application Platform

- Aneka is Manjrasoft Pty. Ltd.'s solution for developing, deploying, and managing cloud applications.
- Aneka consists of a scalable cloud middleware that can be deployed on top of heterogeneous computing resources.
- Aneka includes an extensible set of APIs associated with programming models like MapReduce.
- These APIs support different cloud models like a private, public, hybrid Cloud.

Aneka Framework overview

- Aneka is Manjrasoft Pty. Ltd.'s solution for developing, deploying, and managing cloud applications
- It can be deployed on a variety of computing resources, including computers, servers, data centres, and virtual cloud infrastructures.
- This makes it a versatile and powerful platform for developing and running applications of all sizes and complexity.
- Below figure, provides a complete overview of the components of the Aneka framework.



- An Aneka-based cloud computing is a distributed computing environment that leverages physical and virtualized resources connected through a network.
- Each resource hosts an Aneka Container, which serves as a runtime environment for executing distributed applications and provides essential management features for efficient operation.
- The container features three different classes of services:
  - Fabric Services:
    - These handle infrastructure management.
    - They take care of the underlying resources and infrastructure needed for the cloud.
    - Examples include resource allocation and network management.
  - Foundation Services:
    - These support the core functionality of the Aneka Cloud.
    - They provide essential services that the cloud relies on.
    - Examples include security, data storage, and communication services.
  - Application Services:
    - These are responsible for managing and running applications.
    - They oversee the execution of applications in the Aneka Cloud.
    - Examples include job scheduling and application monitoring.

1. Core Components
Aneka Container: Integral to the Aneka architecture is the Aneka container that forms the core of the environment and is responsible for the management of jobs and tasks across the distributed infrastructure. The middleware hides the fundamental infrastructure and offers a standard API to host applications.

Resource Manager: The resource manager component is another pivotal component that becomes involved in the provisioning and management of the available computational resources in the cloud environment. In turn, it implies deep interactions with the substrate to make decisions on resource provisioning depending on the applications' loads and profiles.

Task Scheduler: The task scheduler component is responsible for managing and scheduling the tasks with the resources available, the dependency they have, and even the performance of the resources and the tasks set. It means to optimize resource application and minimize time as well as cost of the job completion.

- These services are made available to developers and administrators by means of the application management and development layer, which includes interfaces and APIs for developing cloud applications and the management tools and interfaces for controlling Aneka Clouds.
- Aneka implements a service-oriented architecture (SOA), and services are the fundamental components of an Aneka Cloud.
- Services operate at container level and, except for the platform abstraction layer, they provide developers, users, and administrators with all features offered by the framework.

Anatomy of the Aneka container

- The Aneka container is a fundamental component of Aneka Clouds and serves as the runtime environment for services and applications.
- This container, which acts as the deployment unit in Aneka Clouds, is a lightweight software layer created to host services and communicate with the underlying operating system and hardware.
- The services installed in the Aneka container can be classified into three major categories
  - Fabric Services
  - Foundation Services
  - Application Services

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The Platform Abstraction Layer (PAL) in cloud computing acts as an intermediary layer between the cloud infrastructure and the applications running on it. It provides a standardized interface to interact with underlying hardware, operating systems, or virtualized environments. By doing so, PAL abstracts away the complexities of the underlying platform and allows applications to function without being tied to specific hardware or software implementation

1. Platform Abstraction Layer(PAL)

- The Platform Abstraction Layer (PAL) plays a crucial role in Aneka by recognizing the hosting environment and providing the necessary tools for the container's operations. PAL offers several important features:
  - Platform-Independent Implementation: PAL provides a consistent and platform-independent way to interact with the hosting environment.
  - Access to Platform Properties: PAL offers uniform access to the extended and additional properties of the hosting platform.
  - Uniform Access for Remote Nodes: PAL ensures that remote nodes have consistent and platform-independent access.
  - Management Interfaces: PAL provides uniform and platform-independent management interfaces.

Purpose:
Platform Independence: Applications do not need to know about the specifics of the underlying platform (e.g., different cloud providers or environments).
Simplified Development: Developers can focus on building features rather than managing hardware or OS-level differences.

- The Platform Abstraction Layer collects and exposes crucial data from the host system, including information such as the number of CPU cores, CPU frequency, CPU usage, network addresses, memory size, and more.
- This data is made available to the container, enabling it to adapt and function effectively in the hosting environment, regardless of the specific platform.

2. Fabric Services

- Fabric Services define the lowest level of the software stack representing the Aneka Container. They provide access to the resource-provisioning subsystem and to the monitoring facilities implemented in Aneka
- The Fabric Services in Aneka Cloud perform two main functions:
    - Resource monitoring: Fabric services collect and report information about the state of the system's resources, such as CPU usage, memory usage, and network traffic. This information is used by other Aneka services to make informed decisions about resource allocation and scheduling.
    - Resource provisioning: Fabric services provision resources to applications as needed. This includes allocating virtual machines, starting containers, and mounting storage volumes.
    - Resource management: Fabric services manage the lifecycle of resources, including starting, stopping, and restarting them. They also handle resource failures and ensure that applications have the resources they need to run smoothly.

3. Foundation Services
    - These services come into play at a higher level and focus on the logical management of the distributed system that operates on top of the infrastructure created by Fabric Services.
    - They provide additional support for executing distributed applications. Various programming models can integrate with and make use of these services to enhance application management.
    - The services provided by Foundation Service
        - Storage management: Foundation services provide a unified interface for managing storage resources across the Aneka cloud. This includes managing file systems, object storage, and block storage.
        - Accounting, billing, and resource pricing: Foundation services track the usage of resources by Aneka applications and provide mechanisms for accounting, billing, and resource pricing.
        - Resource reservation: Foundation services allow users to reserve resources in advance, ensuring that they have the resources they need when they need them.
        - Security: Foundation services provide a number of security features, such as authentication, authorization, and access control.
        - Monitoring and reporting: Foundation services provide a number of monitoring and reporting services that can be used to track the performance and health of the Aneka cloud.

4. Application services

- Application Services manage the execution of applications and constitute a layer that differentiates according to the specific programming model used for developing distributed applications on top of Aneka.
- The two activities in corresponding services
    - Scheduling Service: - Scheduling Services are in charge of planning the execution of distributed applications on top of Aneka and governing the allocation of jobs composing an application to nodes.

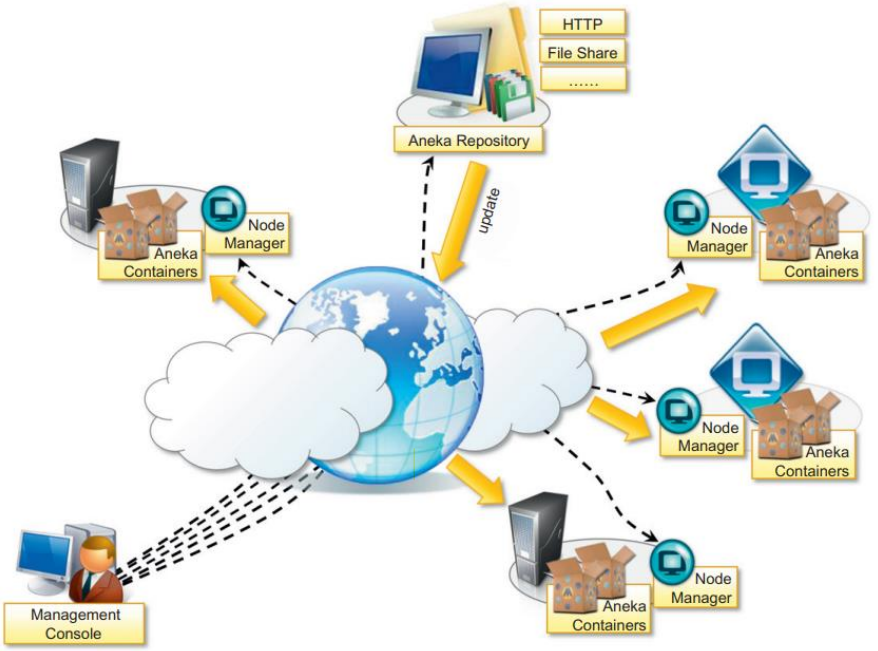
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- Execution Service: - Execution Services control the execution of single jobs that compose applications.
- They are in charge of setting up the runtime environment hosting the execution of jobs.

Building Aneka Cloud

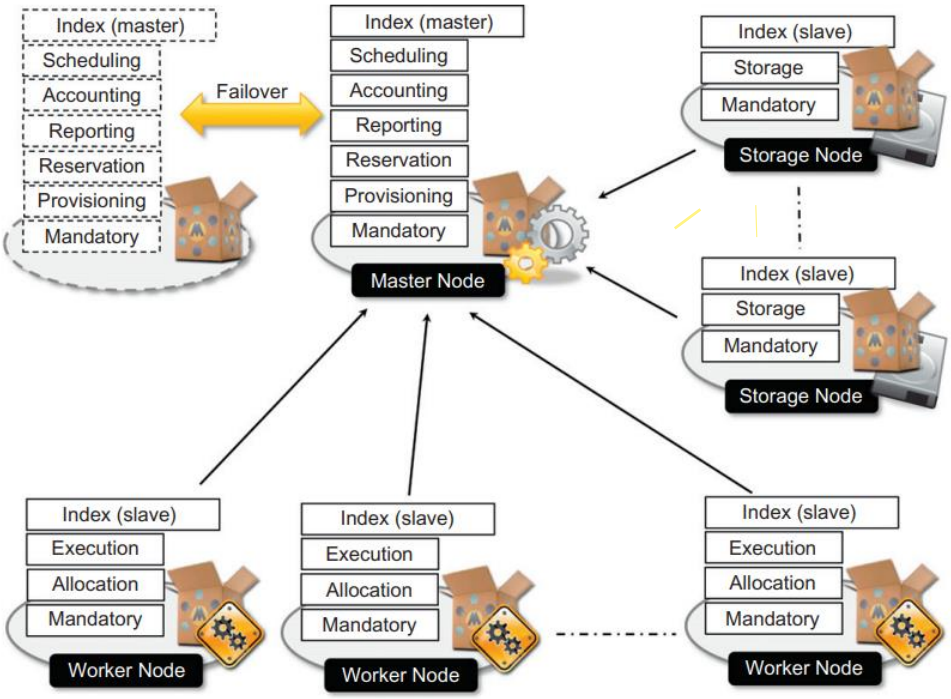
- Aneka is primarily a platform for developing distributed applications for clouds.
- As a software platform it requires infrastructure on which to be deployed; this infrastructure needs to be managed.
  - Infrastructure management tools are specifically designed for this task, and building clouds is one of the primary tasks of administrators.
- Aneka supports various deployment models for public, private, and hybrid clouds.

1. Infrastructure organization
   - The Management Console is the main tool for managing the Aneka Clouds. It can perform various operations, such as creating, configuring, monitoring, and controlling the cloud nodes and containers.
   - The repositories are the storage places for the libraries that are needed to install and run the Aneka platform. The repositories can use different communication channels, such as HTTP, FTP, or file sharing, to make the libraries available.
   - The infrastructure consists of a collection of nodes, each of which runs an Aneka node manager or daemon. These daemons manage the deployment and control of container instances, which are the building blocks of Aneka applications.
   - The nodes can be physical or virtual machines that are connected through a network, such as the internet or a private network. The nodes can have different locations, operating systems, or hardware specifications. The nodes need to have an internet connection and remote administrative access to be part of the Aneka Cloud.
   - The below figure provides an overview of Aneka Clouds from an infrastructure point of view.



Logical Organization

- The logical organization of Aneka Clouds can be very diverse, since it strongly depends on the configuration selected for each of the container instances belonging to the Cloud.
- The most common scenario is to use a master-worker configuration with separate nodes for storage



Logical Organization of an Aneka Cloud

Master node

- The master node is the main node that controls the Aneka Cloud.
- It has the Index Service, which keeps track of all the other nodes and services in the cloud.
- It also has the scheduling services, which assign tasks to the worker nodes.
- It also has a connection to a database, where it stores the state and information of the applications and services.
- The master node can be replicated in several nodes to provide backup and reliability.
- A common configuration of the master node is as follows:
  - Index Service(master copy)
  - Heartbeat Service (monitors node health)
  - Logging Service (collects and stores logs)
  - Reservation Service (manages resource reservations)
  - Resource Provisioning Service (provisions resources for container instances)
  - Accounting Service (tracks resource usage and generates accounting reports)
  - Reporting and Monitoring Service (provides real-time monitoring)
  - Scheduling Services (schedules tasks for execution based on programming model)

Worker node

- The worker nodes are the nodes that execute the applications on the Aneka Cloud.

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- They have the mandatory services, such as the Membership Service, the Message Service, and the Logging Service, which are required for the basic operation and communication of the cloud.
- They also have the execution services, which are specific to each programming model, such as task-based, thread-based, or map-reduce.
- The worker nodes can be divided into different groups, depending on the programming model they support, to balance the load and optimize the performance of the cloud.
- A very common configuration is the following:
  - Index Service
  - Heartbeat Service
  - Logging Service
  - Allocation Service
  - Monitoring Service
  - Execution Services for the supported programming model

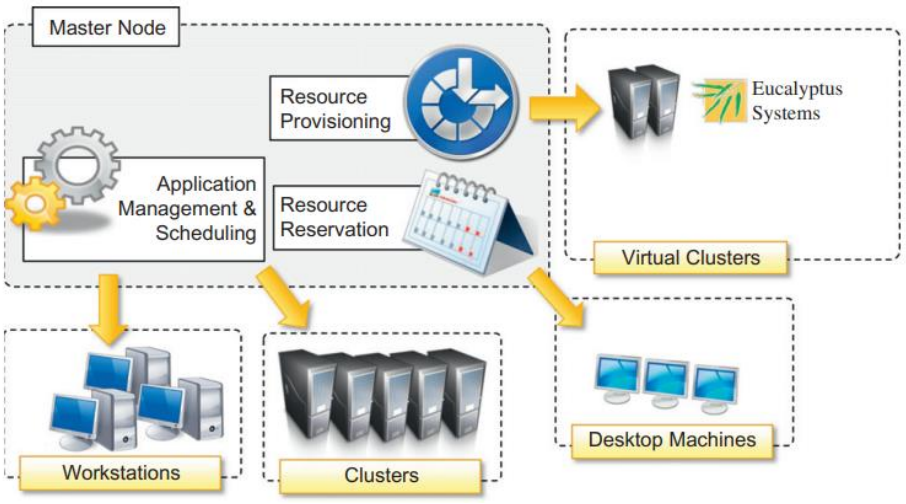
Storage node

- The storage nodes are the nodes that provide storage support to the applications on the Aneka Cloud.
- They have the Storage Service, which allows the applications to store and retrieve files from the cloud.
- They also have the mandatory and usual services, such as the Membership Service, the Message Service, and the Logging Service.
- The common configuration of a storage node is the following:
  - Index Service
  - Heartbeat Service
  - Logging Service
  - Monitoring Service
  - Storage Service

Private cloud deployment mode

- In a private cloud deployment, the Aneka platform is hosted on infrastructure that is dedicated to a single organization.
- This infrastructure may be located on-premises or in a data center, providing the organization with more control over security and customization.

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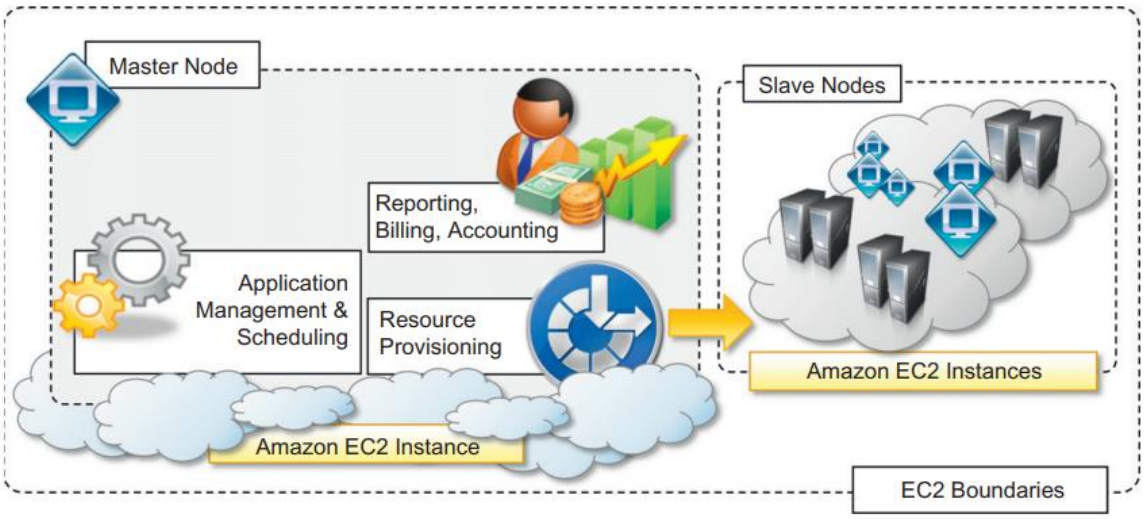


Private cloud deployment

- Private Aneka clouds leverage local physical resources, such as servers, storage devices, and networking components, within the organization's premises.
- Aneka can harness a diverse range of resources, including desktop machines, clusters, and workstations, to create a private cloud.
- The heterogeneous resource pool can be partitioned into different groups based on resource capabilities and application requirements.
- Virtual nodes can be provisioned and integrated into the private cloud using local virtualization platforms like XenServer, Eucalyptus, or OpenStack.
- Private Aneka clouds offer enhanced security and control over the underlying infrastructure, data, and applications.
- This deployment is suitable for scenarios with predictable workloads, where excess capacity demand can be managed using local resource provisioning.

Public Cloud deployment mode

- Public cloud deployment mode features the installation of Aneka master and worker node over a virtualized infrastructure provided by resource provider like Amazon EC2 or GoGrid.
- The below figure shows the pubilc deployment mode



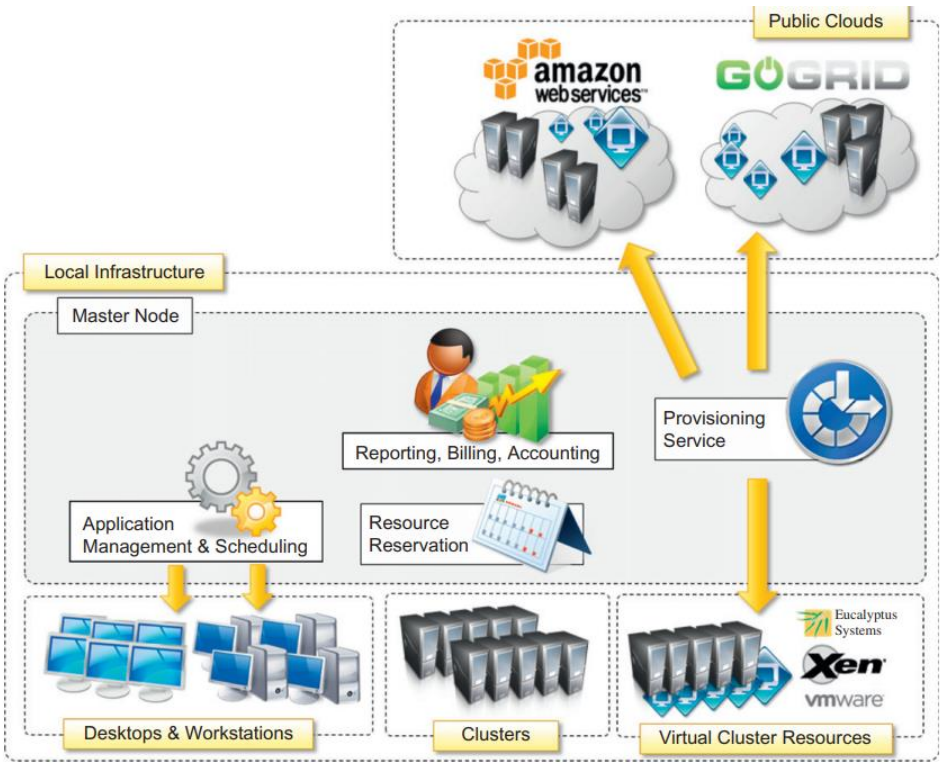
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- There are two ways of deploying Aneka nodes over a public cloud: static deployment and dynamic deployment.
- Static deployment is the method of allocating resources to Aneka nodes before they are deployed and use these nodes as if they were real machines, without any dynamic provisioning capability.
- Dynamic deployment leverages the elasticity features of Infrastructure as a Service (IaaS) providers, allowing the Aneka Cloud infrastructure to scale dynamically based on demand.
- The Resource Provisioning Service plays a crucial role in provisioning dynamic instances based on different images and templates.
- Other important services that have to be included in the master node are the Accounting and Reporting Services. These provide details about resource utilization by users and applications and are fundamental in a multitenant Cloud where users are billed according to their consumption of Cloud capabilities.
- Dynamic instances provisioned on demand will mostly be configured as worker nodes, and, in the specific case of Amazon EC2, different images featuring a different hardware setup can be made available to instantiate worker containers.

- Deployments using different providers are unlikely to happen because of the data transfer costs among providers, but they might be a possible scenario for federated Aneka Clouds. In this scenario resources can be shared or leased among providers under specific agreements and more convenient prices.

Hybrid Cloud deployment mode

- Hybrid deployment model is the most common deployment of Aneka,
- Hybrid cloud is a combination of both private and public clouds, where the existing infrastructure can be elastically scaled on demand.



Hybrid cloud deployment

- Many organizations already possess established computing infrastructures (e.g., on-premises servers, data centers, networks).

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- Aneka is designed to integrate with and make use of the existing computing infrastructure to fulfill application computing needs.
- The existing infrastructure serves as the static deployment of Aneka.
- Despite the static nature, Aneka provides the capability for elastic scaling.
- Elastic scaling allows for dynamic and flexible increases in resources when there's a demand for additional computing resources.
- Elastic scaling on demand ensures optimal resource utilization.
- Strikes a balance by efficiently using the existing infrastructure while being adaptable to changing workloads.
- Aneka can coordinate the execution of applications across different clouds, using different services, such as the Resource Provisioning Service, the Accounting Service, and the Reporting Service

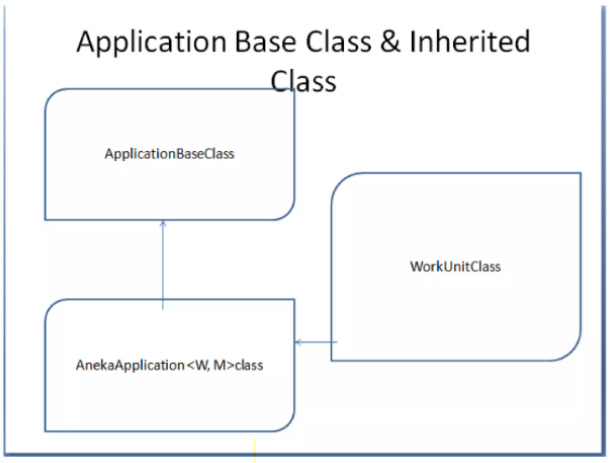
Cloud programming and management

- Aneka's primary purpose is to provide a scalable middleware product in which to execute distributed applications.

- Application development and management constitute the two major features that are exposed to developers and system administrators.
- To simplify these activities, Aneka provides developers with a comprehensive and extensible set of APIs and administrators with powerful and intuitive management tools.
- The APIs for development are mostly concentrated in the Aneka SDK and management tools are exposed through the Management Console.

1. Aneka SDK
   - Aneka provides APIs for developing applications on top of existing programming models, implementing new programming models, and developing new services to integrate into the Aneka Cloud.
   - The SDK provides support for both programming models and services by means of the Application Model and the Service Model.

a. Application model
   - The application model covers the development of applications and new programming models.
   - It consists of application class and application manager
     a. Application class: - Provides user/developer view about distributed applications of Aneka cloud.
     b. Application Manager:- Are internal components that control and monitor the execution of aneka clouds.
   - Each distributed application running on top of Aneka is an instance of the ApplicationBase <M>. class, where M identifies the specific type of application manager used to control the application.



Interface C tions (ICCA)

   - Aneka further specializes applications into two main categories:
     (1) Applications whose tasks are generated by the user
     (2) Applications whose tasks are generated by the runtime infrastructure.

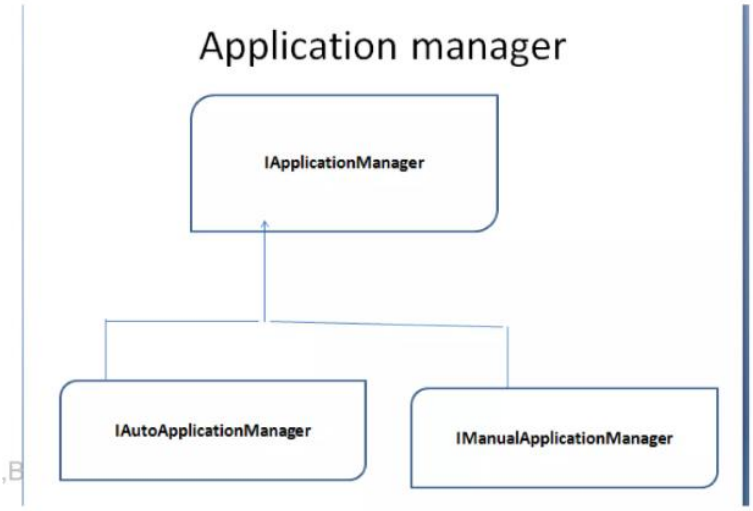
Applications whose tasks are generated by the user

- The first category is the most common and it is used as a reference for several programming models supported by Aneka: the Task Model, the Thread Model, and the Parameter Sweep Model.
- Applications that fall into this category are composed of a collection of units of work submitted by the user and represented by the WorkUnit class
- All the applications that fall into this category are instances of AnekaApplication <W,M >., where W is the specific type of WorkUnit class

used, and M is the type of application manager used to implement the IManualApplicationManager interface.

Applications whose tasks are generated by the runtime infrastructure

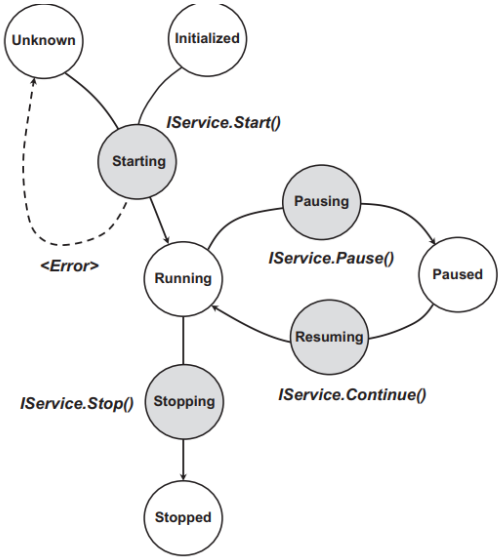
- The second category covers the case of MapReduce and all those other scenarios in which the units of work are generated by the runtime infrastructure rather than the user.
- In this case there is no common unit-of-work class used, and the specific classes used by application.
- All the applications that fall into this category is are instance of ApplicationBase <M>, where M implements IAutoApplicationManager.



b. Service Model

- The Service Model defines the general infrastructure for service development.
- The Aneka Service Model defines the basic requirements to implement a service that can be hosted in an Aneka Cloud.
- The container defines the runtime environment in which services are hosted.
- Each service that is hosted in the container must use IServiceinterface, which exposes the following methods and properties:
  - Name and status
  - Control operations such as Start, Stop, Pause, and Continue methods
  - Message handling by means of the HandleMessage method
- Below figure describes the reference life cycle of each service instance in the Aneka container.

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- A service instance can initially be in the Unknown or Initialized state, a condition that refers to the creation of the service instance by invoking its constructor during the configuration of the container.
- Once the container is started, it will iteratively call the Start method on each service method.
- As a result the service instance is expected to be in a Starting state until the startup process is complete&, after which it will exhibit the Running state.
- If an exception occurs while starting the service, it is expected that the
- service will fall back to the Unknown state, thus signaling an error.
- Running state is the condition in which the service will last as long as the container is active and running. This is the only state in which the service is able to process messages.
- When a service is running it is possible to pause its activity by calling
- the Pause method and resume it by calling Continue.
- When the container shutdown, the Stop method is iteratively called
- on each service running, and services move first into the transient
- Stopping state to reach the final Stopped state, where all resources that were initially allocated have been released.

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2. Management tools
    - Aneka is a pure PaaS implementation and requires virtual or physical hardware to be deployed.
    - The management layer of Aneka is responsible for managing this infrastructure, including provisioning and managing virtual machines, installing and configuring Aneka components, and monitoring the health and performance of the cloud.
    - In addition to infrastructure management, the management layer also provides capabilities for managing services and applications running in the Aneka Cloud.

    a. Infrastructure management
        - Aneka leverages virtual and physical hardware in order to deploy Aneka Clouds.
        - Virtual hardware is generally managed by means of the Resource Provisioning Service, which acquires resources on demand according to the need of applications, while physical hardware is directly

managed by the Administrative Console by using the Aneka management API of the PAL.

b. Platform management

- A collection of connected containers defines the platform on top of which applications are executed.
- The features available for platform management are mostly concerned with the logical organization and structure of Aneka Clouds.
- It is possible to partition the available hardware into several Clouds variably configured for different purposes.
- Services implement the core features of Aneka Clouds and the management layer exposes operations for some of them, such as Cloud monitoring, resource provisioning and reservation, user management, and application profiling.

c. Application management

- Applications identify the user contribution to the Cloud.
- The management APIs provide administrators with monitoring and profiling features that help them track the usage of resources and relate them to users and applications.
- This is an important feature in a cloud computing scenario in which users are billed for their resource usage.
- Aneka exposes capabilities for giving summary and detailed information about application execution and resource utilization.
- All these features are made accessible through the Aneka Cloud Management Studio, which constitutes the main Administrative Console for the Cloud.

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Unit IV : Cloud Platforms in Industry

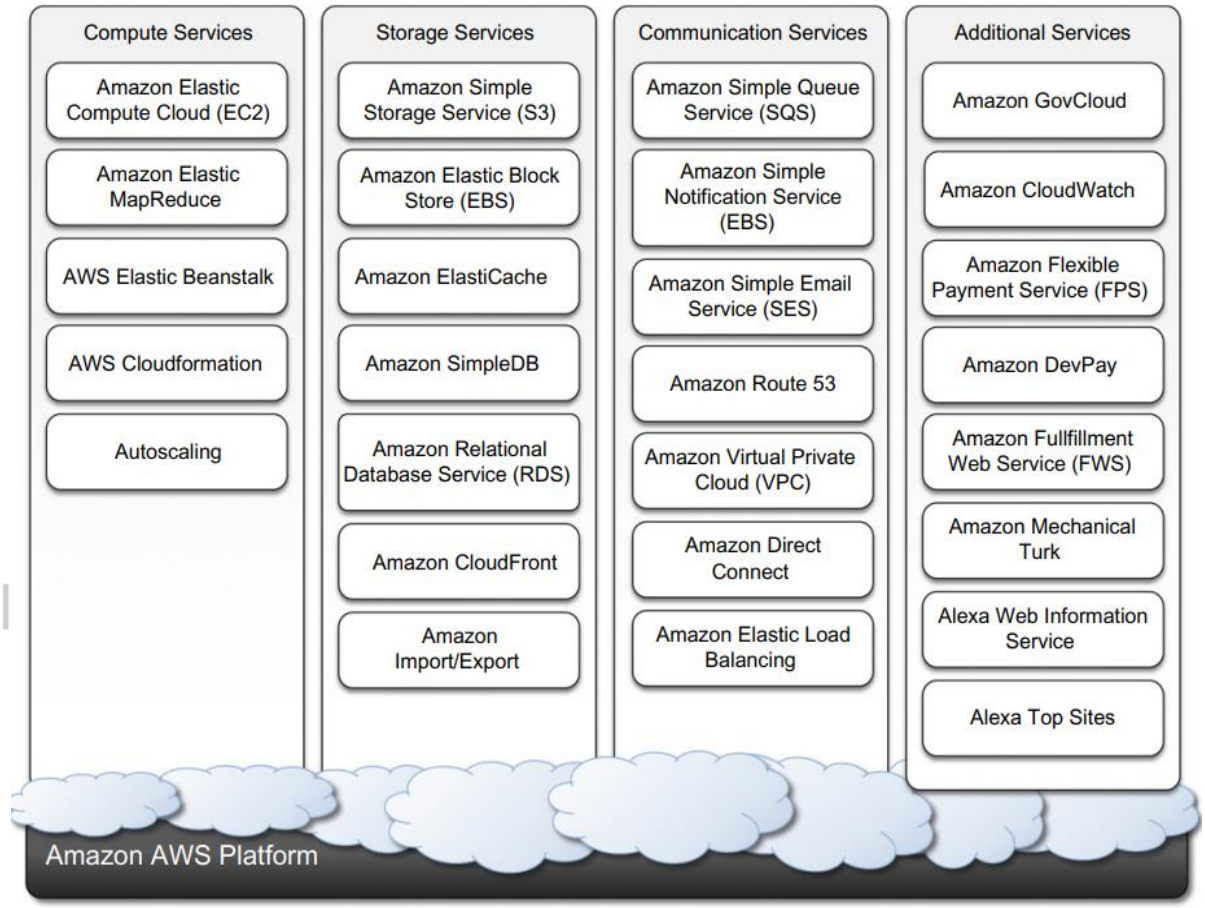
Amazon Web Services(AWS):

- AWS is a platform that helps you create flexible applications that can run on the cloud, which is a network of computers that provide resources and services for your applications.
- AWS offers solutions for elastic infrastructure scalability, messaging, and data storage, which means that you can easily adjust the amount and type of resources and services you need for your applications, depending on the workload and demand.

AWS (Amazon Web Services) is a comprehensive cloud computing platform provided by Amazon. It offers a wide range of on-demand cloud services, including computing power, storage, databases, machine learning, and more, on a pay-as-you-go pricing model

Services provided by AWS

- Computer services
- Storage service
- Communication service
- Additional service



Amazon Web Services ecosystem.

Compute Services:

- Compute services are a type of cloud computing service that provide the infrastructure and resources for running applications and workloads on the cloud.
- Compute services allow users to create, manage, and scale virtual machines, containers, functions, and other compute resources on demand, without having to own or maintain the physical hardware.

Amazon Elastic Compute Cloud (Amazon EC2)

- Amazon EC2 is a service that provides an IaaS solution for cloud computing, which means that it allows you to create and run virtual servers on the cloud, using the resources and services of Amazon.

i. Amazon machine image(AMI)

- Amazon EC2 lets you deploy servers as virtual machines, which are called instances. Instances are created from images, which are templates that have a preinstalled operating system and a software stack. You can choose from different types of images, depending on your needs and preferences.

ii. EC2 Instances

- EC2 instances represent virtual machines.
- They are created using AMI as templates, which are specialized by selecting the number of cores, their computing power, and the installed memory.
- Instances available in EC2
    - Standard instances. This class offers a set of configurations that are suitable for most applications.
    - Micro instances. This class is suitable for those applications that consume a limited amount of computing power and memory and occasionally need bursts in CPU cycles
    - High-memory instances. This class targets applications that need to process huge workloads and require large amounts of memory
    - High-CPU instances. This class targets compute-intensive applications.
    - Cluster Compute instances. This class is used to provide virtual cluster services. Instances in this category are characterized by high CPU compute power and large memory and an extremely high I/O and network performance.
    - Cluster GPU instances. This class provides instances featuring graphic processing units (GPUs) and high compute power, large memory, and extremely high I/O and network performance. This class is particularly suited for cluster applications that perform heavy graphic computations.

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EC2 Environment

- The EC2 environment is a virtual environment that provides the services and resources for running EC2 instances, which are virtual machines that host applications on the cloud.
- The EC2 environment is responsible for allocating addresses, attaching storage volumes, and configuring security for the EC2 instances. These services help the instances to communicate and access data on the cloud.
- By default, EC2 instances have an internal IP address, which allows them to communicate within the EC2 network and access the Internet as clients. However, users can also associate an Elastic IP to each instance, which is a public IP address that can be remapped to a different instance over time. This can help users to have more control and flexibility over their instances.

Advanced compute services

1. **AWS CloudFormation**
   - AWS CloudFormation is a service that helps you model and set up your AWS resources so that you can spend less time managing those resources and more time focusing on your applications that run in AWS.
   - You create a template that describes all the AWS resources that you want (like Amazon EC2 instances or Amazon RDS DB instances), and CloudFormation takes care of provisioning and configuring those resources for you.
   - You don't need to individually create and configure AWS resources and figure out what's dependent on what; CloudFormation handles that.

2. **Amazon Elastic Beanstalk**
   - AWS Elastic Beanstalk helps to quickly deploy and manage applications in the AWS Cloud without having to worry about the infrastructure that runs those applications.
   - Reduces management complexity without restricting choice or control.
   - Elastic Beanstalk automatically launches an environment once an application is uploaded, and creates and configures the AWS resources needed to run the code. After the environment is launched, it can be managed and used to deploy new application versions.

3. **Amazon Elastic MapReduce**
   - Amazon Elastic MapReduce (EMR) is a service that helps you create and manage big data applications on the AWS cloud.
   - You can use EMR to run frameworks such as Apache Hadoop, Apache Spark, and Presto, which can process and analyze large amounts of data.
   - EMR also lets you integrate your applications with other AWS services, such as Amazon S3, Amazon DynamoDB, and Amazon SageMaker.can help perform data-intensive tasks for applications such as web indexing, data mining, log file analysis, machine learning, financial analysis, scientific simulation, bioinformatics research, etc

Storage Services:

- AWS provides a collection of services for data storage and information management. The core service in this area is represented by Amazon Simple Storage Service (S3).

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Amazon Simple Storage Services(S3)

- AWS provides a collection of services for data storage and information management.
- The core service in this area is represented by Amazon Simple Storage Service (S3).
- This is a distributed object store that allows users to store information in different formats.
- The core components of S3 are two: buckets and objects.
  1. Bucket: -
     - Buckets represent virtual containers in which to store objects
     - A bucket is a container of objects.
     - Buckets are top level elements of the S3 storage architecture and do not support nesting.
     - That is, it is not possible to create “subbuckets” or other kinds of physical divisions.

- Users create a bucket by sending a PUT request to http://s3.amazonaws.com/ with the name of the bucket.
- The content of a bucket can be listed by sending a GET request specifying the name of the bucket.
- Once created, the bucket cannot be renamed or relocated.

2. Object:-
   - Objects represent the content that is actually stored.
   - Objects can also be enriched with metadata that can be used to tag the stored content with additional information.
   - An object is identified by a name that needs to be unique within the bucket in which the content is stored.
   - Users create an object via a PUT request that specifies the name of the object together with the bucket name, its contents, and additional properties. T
   - he maximum size of an object is 5 GB.
   - Once an object is created, it cannot be modified, renamed, or moved into another bucket.
   - It is possible to retrieve an object via a GET request; deleting an object is performed via a DELETE request.

a. S3 key concepts

- S3 has been designed to provide a simple storage service that's accessible through a Representational State Transfer (REST) interface, which is quite similar to a distributed file system but which presents some important differences that allow the infrastructure to be highly efficient:
- The difference between distributed file system and S3
  - The storage is organized in a two-level hierarchy. S3 organizes its storage space into buckets that cannot be further partitioned.
  - Stored objects cannot be manipulated like standard files. S3 has been designed to essentially provide storage for objects that will not change over time. Therefore, it does not allow renaming, modifying, or relocating an object
  - Content is not immediately available to users :- It is a large distributed storage facility, changes are not immediately reflected.
  - Requests will occasionally fail. Due to the large distributed infrastructure being managed, requests for object may occasionally fail.

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b. Resource naming in S3

- Amazon offers three different ways of addressing a bucket:
  - Canonical form: http://s3.amazonaws.com/bukect_name/. The bucket name is expressed as a path component of the domain name s3.amazonaws.com.
  - Subdomain form: http://bucketname.s3.amazon.com/. Alternatively, it is also possible to reference a bucket as a subdomain of s3.amazonaws.com.
  - Virtual hosting form: http://bucket-name.com/. Amazon also allows referencing of its resources with custom URLs. This is accomplished

by entering a CNAME record into the DNS that points to the subdomain form of the bucket URI.

c. Access control and security

- Amazon S3 allows controlling the access to buckets and objects by means of Access Control Policies (ACPs).
- An ACP is a set of grant permissions that are attached to a resource expressed by means of an XML configuration file.
- Five different permissions can be used:
  - READ allows the grantee to retrieve an object and its metadata and to list the content of a bucket as well as getting its metadata.
  - WRITE allows the grantee to add an object to a bucket as well as modify and remove it.
  - READ_ACP allows the grantee to read the ACP of a resource.
  - WRITE_ACP allows the grantee to modify the ACP of a resource.
  - FULL_CONTROL grants all of the preceding permissions.

Other Amazon storage services

1. **Amazon Elastic Block Store(EBS)**
   - Amazon Elastic Block Store (EBS) is a service that provides block-level storage volumes for use with Amazon EC2 instances.
   - Block-level storage volumes are like raw, unformatted hard drives that you can attach to your instances and use as you wish.
   - You can create a file system on them, or use them for any applications that need direct access to the storage device.
   - Amazon EBS volumes are persistent, which means they remain even after you stop or terminate your instances.
   - You can also detach them from one instance and attach them to another instance, as long as they are in the same Availability Zone.
   - Amazon EBS volumes are also scalable, which means you can adjust their size, performance, and type according to your needs.

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2. **Amazon ElasticCache**
   - Amazon ElastiCache is a service that helps you create and manage in-memory cache environments on the AWS cloud.
   - It supports two popular open-source caching engines: Redis and Memcached.
   - Use ElastiCache to improve the performance and scalability of your web applications by storing frequently accessed data in memory, reducing the load on your databases and servers.
   - You can also use ElastiCache to store and process data for real-time applications, such as gaming, e-commerce, social media, and analytics.
   - ElastiCache is a fully managed service, which means that AWS takes care of the provisioning, configuration, monitoring, and maintenance of your cache clusters.
   - You can also use ElastiCache to integrate with other AWS services, such as Amazon S3, Amazon DynamoDB, and Amazon SageMaker

3. **Amazon SimpleDB**
   - Amazon SimpleDB is a lightweight, highly scalable, and flexible data storage solution for applications that do not require a fully relational model for their data.

- SimpleDB provides support for semistructured data, the model for which is based on the concept of domains, items, and attributes.
- With respect to the relational model, this model provides fewer constraints on the structure of data entries, thus obtaining improved performance in querying large quantities of data.
- As happens for Amazon RDS, this service frees AWS users from performing configuration, management, and high-availability design for their data stores.

4. **Amazon Relational Database Service(RDS)**
   - Amazon RDS is a service that helps you create and manage relational databases on the AWS cloud.
   - You can use Amazon RDS to run different database engines, such as MySQL, PostgreSQL, Oracle, SQL Server, and Amazon Aurora. Amazon RDS takes care of the common database administration tasks, such as backup, recovery, patching, scaling, and security.

5. **Amazon CloudFront:**
   - CloudFront is a fully managed, fast content delivery network (CDN) service that speeds up the distribution of static, dynamic web, or streaming content to end-users.
   - CloudFront delivers the content through a worldwide network of data centers called edge locations or Point of Presence (POP).
   - CloudFront securely delivers data, videos, applications, and APIs to customers globally with low latency, and high transfer speeds, all within a developer-friendly environment.

Communication services

- Amazon provides facilities to structure and facilitate the communication among existing applications and services residing within the AWS infrastructure. These facilities can be organized into two major categories: virtual networking and messaging.

1. **Virtual Networking**
   - Virtual networking Virtual networking comprises a collection of services that allow AWS users to control the connectivity to and between compute and storage services.
   - Amazon Virtual Private Cloud (VPC) and amazon Direct Connect provide connectivity solutions in terms of infrastructure; Route 53 facilitates connectivity in terms of naming

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i. Amazon Virtual Private Cloud(VPC)

- Amazon VPC is a web service that allows you to create and manage your own virtual network on the cloud.
- We can use Amazon VPC to launch AWS resources, such as EC2 instances and RDS databases, in a logically isolated environment that you define.
- We can also customize your VPC with features such as subnets, IP addresses, gateways, routing, security, and connectivity.

ii. Amazon Direct Connect

- Amazon Direct Connect allows AWS users to create dedicated networks between the user private network and Amazon Direct Connect locations, called ports.
- This connection can be further partitioned in multiple logical connections and give access to the public resources hosted on the Amazon infrastructure.

- The advantage of using Direct Connect versus other solutions is the consistent performance of the connection between the users' premises and the Direct Connect locations.

iii. Amazon Route 53

- Amazon Route 53 is a highly available and scalable Domain Name System (DNS) web service. Route 53 connects user requests to internet applications running on AWS or on-premises. You can use Route 53 to perform three main functions in any combination: domain registration, DNS routing, and health checking.

2. Messaging

- The three different types of messaging services offered are Amazon Simple Queue Service (SQS), Amazon Simple Notification Service (SNS), and Amazon Simple Email Service (SES).

i. Amazon simple queue services:

- Amazon SQS is a service that allows applications to exchange messages through message queues hosted on AWS.
- Amazon SQS enables users to create and configure unlimited message queues and control their access.
- Amazon SQS stores messages securely and redundantly on AWS for a limited time and allows authorized applications to access them.
- Amazon SQS provides a disconnected model of communication between applications that do not need to be online at the same time.

ii. Amazon Simple Notification Services(SNS):

- Amazon Simple Notification Service (Amazon SNS) is a managed service that provides message delivery from publishers to subscribers (also known as producers and consumers) .
- Publishers communicate asynchronously with subscribers by sending messages to a topic, which is a logical access point and communication channel.
- Clients can subscribe to the SNS topic and receive published messages using a supported endpoint.
- The service provides subscribers with different notification models (HTTP/HTTPS, email/email JSON, and SQS)

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iii. Amazon Simple Email Service(SES)

- Amazon SES provides AWS users with a scalable email service that leverages the AWS infrastructure. Once users are signed up for the service, they have to provide an email that SES will use to send emails on their behalf

Additional services

- Besides compute, storage, and communication services, AWS provides a collection of services that allow users to utilize services in aggregation. The two relevant services are Amazon CloudWatch and Amazon Flexible Payment Service (FPS).

i. Amazon CloudWatch

- It is a free service that helps you understand and optimize your applications running on AWS.
- It collects data from various AWS services, such as EC2, S3, SimpleDB, and CloudFront, and provides you with detailed insights into your usage patterns and costs.
- You can use this information to improve the efficiency and cost-effectiveness of your applications.

ii. Amazon Flexible Payment

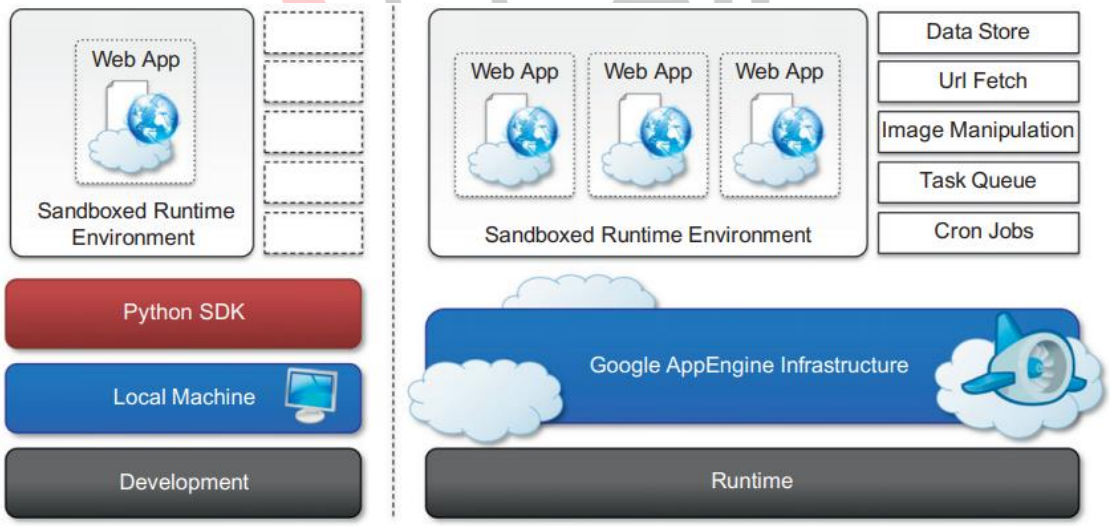
- Amazon FPS infrastructure allows AWS users to leverage Amazon's billing infrastructure to sell goods and services to other AWS users.
- Using Amazon FPS, developers do not have to set up alternative payment methods, and they can charge users via a billing service.
- The payment models available through FPS include one-time payments and delayed and periodic payments, required by subscriptions and usage-based services, transactions, and aggregate multiple payments.

Google AppEngine:-

- Google AppEngine is a PaaS implementation that provides services for developing and hosting scalable Web applications.

Architecture and core concepts

- AppEngine is a platform for developing scalable applications accessible through the Web.
- The platform is logically divided into four major components: infrastructure, the run time environment, the underlying storage, and the set of scalable services that can be used to develop applications



GoogleApp Engine platform architecture

- **Infrastructure**
  - AppEngine hosts Web applications, and its primary function is to serve users requests efficiently.
  - To do so, AppEngine's infrastructure takes advantage of many servers available within Google datacenters.

- For each HTTP request, AppEngine locates the servers hosting the application that processes the request, evaluates their load, and, if necessary allocates additional resources (i.e., servers) or redirects the request to an existing server.
- The infrastructure is also responsible for monitoring application performance and collecting statistics on which the billing is calculated.

- **Runtime environment**
  - The runtime environment represents the execution context of applications hosted on AppEngine.
    a. Sandboxing
       One of the major responsibilities of the runtime environment is to provide the application environment with an isolated and protected context in which it can execute without causing a threat to the server and without being influenced by other applications.
    b. Supported runtimes
       Currently, it is possible to develop AppEngine applications using three different languages and related technologies: Java, Python, and Go.
- **Storage**
  - AppEngine provides various types of storage, which operate differently depending on the volatility of the data.
  - There are three different levels of storage: in memory-cache, storage for semistructured data, and long-term storage for static data
    a. Static file servers
       - Web applications are composed of dynamic and static data.
       - Dynamic data are a result of the logic of the application and the interaction with the user.
       - Static data often are mostly constituted of the components that define the graphical layout of the application (CSS files, plain HTML files, JavaScript files, images, icons, and sound files) or data files.
       - These files can be hosted on static file servers, since they are not frequently modified.

    b. DataStore
       - DataStore is a service that allows developers to store semistructured data.
       - The service is designed to scale and optimized to quickly access data.
       - DataStore can be considered as a large object database in which to store objects that can be retrieved by a specified key.

    c. MemCache
       i. AppEngine provides caching services by means of MemCache. This is a distributed in-memory cache that is optimized for fast access and provides developers with a volatile store for the objects that are frequently accessed.
- **Scalable Services**

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- There are two types of services are available:- Application Service and Compute Service

a. Application Service
   - Applications hosted on AppEngine take the most from the services made available through the runtime environment.
   - These services simplify most of the common operations that are performed in Web applications: access to data, account management, integration of external resources, messaging and communication, image manipulation, and asynchronous computation.

   i. UrlFetch
      - The URL Fetch service in Google App Engine is a service that allows your application to make HTTP/HTTPS requests to external URLs.

   ii. MemCache
      - AppEngine provides caching services by means of MemCache. This is a distributed in-memory cache that is optimized for fast access and provides developers with a volatile store for the objects that are frequently accessed.

   iii. Mail and instant messaging
      - AppEngine provides developers with the ability to send and receive mails through Mail. The service allows sending email on behalf of the application to specific user accounts. It is also possible to include several types of attachments and to target multiple recipients.
      - AppEngine provides also another way to communicate with the external world: the Extensible Messaging and Presence Protocol (XMPP). Any chat service that supports XMPP, such as Google Talk, can send and receive chat messages to and from the Web application.

   iv. Account management
      - Web applications often keep various data that customize their interaction with users.
      - These data normally go under the user profile and are attached to an account.
      - AppEngine simplifies account management by allowing developers to leverage Google account management by means of Google Accounts.

   v. Image manipulation
      - Web applications render pages with graphics. Often simple operations, such as adding watermarks or applying simple filters, are required.
      - AppEngine allows applications to perform image resizing, rotation, mirroring, and enhancement by means of Image Manipulation.

b. Compute services:-
   - Web applications are mostly designed to interface applications with users by means of a ubiquitous channel, that is, the Web.

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- Most of the interaction is performed synchronously: Users navigate the Web pages and get instantaneous feedback in response to their actions.
- This feedback is often the result of some computation happening on the Web application, which implements the intended logic to serve the user request.
- AppEngine offers additional services such as Task Queues and Cron Jobs that simplify the execution of computations
  i. Task queues
    - Task Queues allow applications to submit a task for a later execution. This service is particularly useful for long computations that cannot be completed within the maximum response time of a request handler.
  ii. Cron jobs
    - Sometimes the length of computation might not be the primary reason that an operation is not performed within the scope of the Web request. It might be possible that the required operation needs to be performed at a specific time of the day.

Application life cycle:-

AppEngine provides support for almost all the phases characterizing the life cycle of an application: testing and development, deployment, and monitoring. The SDKs released by Google provide developers with most of the functionalities required by these tasks. Currently there are two SDKs available for development: Java SDK and Python SDK.

1. Application development and testing

- Developers can start building their Web applications on a local development server. This is a self-contained environment that helps developers tune applications without uploading them to AppEngine.
- The development server simulates the AppEngine runtime environment by providing a mock implementation of DataStore, MemCache, UrlFetch, and the other services leveraged by Web applications.

The SDK supported by AppEngine

a. Java SDK

- The Java SDK provides developers with the facility for building applications with the Java 5 and Java 6 runtime environments. Alternatively, it is possible to develop applications within the Eclipse development environment by using the Google AppEngine plug-in, which integrates the features of the SDK within the powerful Eclipse environment.

b. Python SDK

- The Python SDK allows developing Web applications for AppEngine with Python 2.5. It provides a standalone tool, called GoogleAppEngineLauncher, for managing Web applications locally and deploying them to AppEngine.

2. Application deployment and management

- The application has been developed and tested, it can be deployed on AppEngine with a simple click or command-line tool.
- Before performing such task, it is necessary to create an application identifier, which will be used to locate the application from the Web browser by typing the address http://<application-id>.appspot.com.

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- Once an application identifier has been created, it is possible to deploy an application on AppEngine.
- This task can be done using either the respective development environment (GoogleAppEngineLauncher and Google AppEngine plug-in) or the command-line tools.
- Developers can then manage the application by using the administrative console. This is the primary tool used for application monitoring and provides users with insight into resource usage (CPU, bandwidth) and services and other useful counters.

Cost model

- AppEngine provides a free service with limited quotas that get reset every 24 hours.
- Once the application has been tested and tuned for AppEngine, it is possible to set up a billing account and obtain more allowance and be charged on a pay-per-use basis.
- This allows developers to identify the appropriate daily budget that they want to allocate for a given application.
- Project owners and billing administrators can enable billing for a project.

Quotas

- A quota restricts how much of a particular shared Google Cloud resource your Google Cloud project can use, including hardware, software, and network components.

Types of quotas

The following types of quotas apply to App Engine applications:

1. Free quotas give your application an amount of each resource for free. After your application exceeds a free quota, you will be billed for any additional use of that resource.
   Only the App Engine standard environment provides free quotas.
2. Daily quotas protect the integrity of the App Engine system by ensuring that no single app over-consumes a resource to the detriment of other apps. If you go above these limits, you'll get an error. Daily quotas are refreshed daily at midnight Pacific time.
3. Per-minute quotas protect your application from consuming all of its resources in very short periods of time, and prevent other applications from monopolizing a given resource.

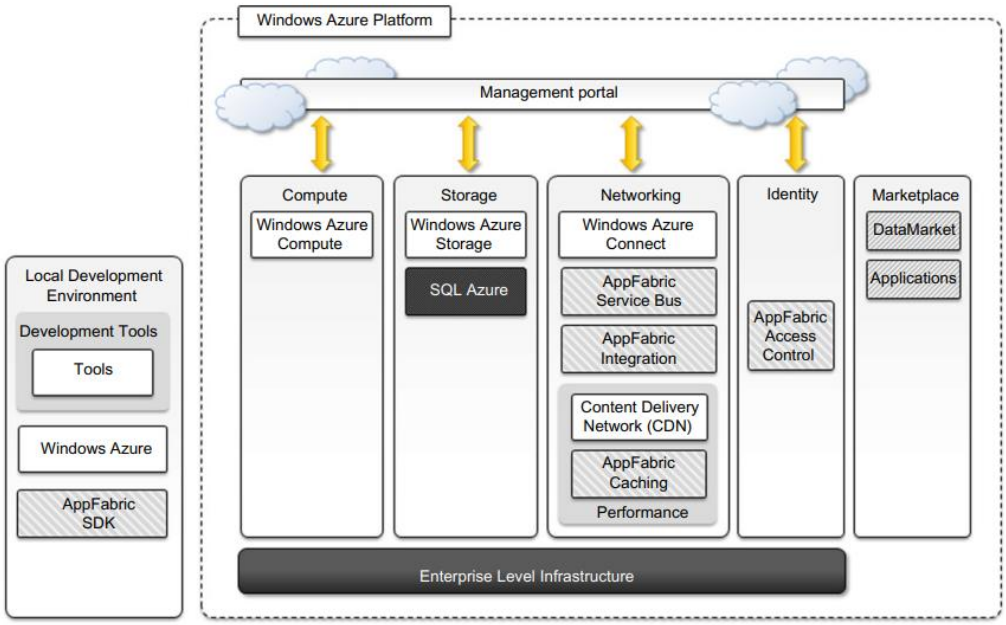
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Observations

- Google App Engine enables you to build web applications for your business leveraging Google's infrastructure. App Engine applications are easy to develop, maintain, and can scale as your traffic and data storage needs grow.

Azure core concepts

- The Windows Azure platform is made up of a foundation layer and a set of developer services that can be used to build scalable applications.
- These services cover compute, storage, networking, and identity management, which are tied together by middleware called AppFabric.
- This scalable computing environment is hosted within Microsoft data centers and accessible through the Windows Azure Management Portal.



1. **Compute services** Azure Compute Services are a key offering from Microsoft Azure, providing the computational power needed to run applications, manage workloads, and process data in the cloud. Azure Compute enables users to deploy, run, and manage applications on-demand
   - Compute services are the core components of Microsoft Windows Azure, and roles are the abstraction that allows users to consume these services.
   - Currently, there are three different roles: Web role, Worker role, and Virtual Machine (VM) role.
     - Web roles are designed to host web applications and APIs. They provide a fully managed environment that includes a web server, application pool, and database.
     - Worker roles are designed to run background tasks, such as batch processing and job scheduling. They provide a flexible environment that can be customized to meet the specific needs of your application.
     - Virtual Machine (VM) roles provide the greatest flexibility, allowing you to run any type of operating system or application on the Azure platform.

- Roles are managed by the Azure operating system, which allows them to be scaled up or down on demand to meet the needs of your application.
- To use roles, you simply need to define the roles that you need for your application and then deploy your application to Azure. Azure will then create and manage the roles on your behalf.
- If the traffic to your web application increases, Azure will automatically create new web role instances to handle the additional traffic. This allows your web application to scale up seamlessly to meet the demands of your users.
- Roles are a powerful abstraction that allows users to easily consume the compute services that are provided by Windows Azure. They provide a flexible and scalable way to run any type of application on the Azure platform.

Azure Virtual Machines (VMs)
What it is: Scalable, on-demand virtual servers in the cloud
Azure Functions: Serverless compute service to run code without managing servers.

Storage services Azure Storage Services provide scalable, secure, and durable storage options for a variety of data types. These services are foundational to many applications, offering solutions for structured, unstructured, and semi-structured data.

- Windows Azure provides different types of storage solutions that complement compute services with a more durable and redundant option compared to local storage.
- Compared to local storage, these services can be accessed by multiple clients at the same time and from everywhere, thus becoming a general solution for storage

1. Blobs Object storage for unstructured data like documents, videos, images, and backups.

- Azure allows storing large amount of data in the form of binary large objects (BLOBs) by means of the blobs service.
- This service is optimal to store large text or binary files. Two types of blobs are available:
  - Block blobs. Block blobs are composed of blocks and are optimized for sequential access; therefore they are appropriate for media streaming.
  - Page blobs. Page blobs are made of pages that are identified by an offset from the beginning of the blob. A page blob can be split into multiple pages or constituted of a single page. This type of blob is optimized for random access and can be used to host data different from streaming.
- Blobs storage provides users with the ability to describe the data by adding metadata. It is also possible to take snapshots of a blob for backup purposes.

2. Azure drive High-performance, durable block storage for virtual machines.

- Azure Disk Storage is a block-level storage service that provides high-performance, durable storage for Azure Virtual Machines.
- Azure Disks are managed by Azure, so you don't have to worry about provisioning, managing, or maintaining storage.

3. Tables A NoSQL datastore for structured, non-relational data.

- Tables constitute a semistructured storage solution, allowing users to store information in the form of entities with a collection of properties.
- Entities are stored as rows in the table and are identified by a key, which also constitutes the unique index built for the table.
- Users can insert, update, delete, and select a subset of the rows stored in the table.

4. Queues service for storing and managing large numbers of messages in a queue.

- Queue storage allows applications to communicate by exchanging messages through durable queues, thus avoiding lost or unprocessed messages.
- Applications enter messages into a queue, and other applications can read them in a first-in, first-out (FIFO) style

Core infrastructure: AppFabric

- AppFabric is a comprehensive middleware for developing, deploying, and managing applications on the cloud or for integrating existing applications with cloud services.
- AppFabric implements an optimized infrastructure supporting scaling out and high availability; sandboxing and multitenancy; state management; and dynamic address resolution and routing.

AppFabric provides a number of services, including:

1. AppFabric Caching: A distributed, in-memory cache that can be used to improve the performance and scalability of applications. Provided distributed in-memory caching to improve application performance.
2. AppFabric Service Bus: A messaging service that can be used to connect applications and services together. A messaging infrastructure for distributed systems. Provided reliable message delivery between applications or components running on-premises or in Azure.
3. AppFabric Access Control: A service that can be used to manage the identity and access control for applications. Enabled authentication and authorization for cloud applications.
4. AppFabric Integration: A service that can be used to integrate applications with on-premises and cloud-based systems.

Other services

- Compute, storage, and middleware services constitute the core components of the Windows Azure platform.

- Besides these, other services and components simplify the development and integration of applications with the Azure Cloud.
- An important area for these services is applications connectivity, including virtual networking and content delivery.

Windows Azure virtual network :-

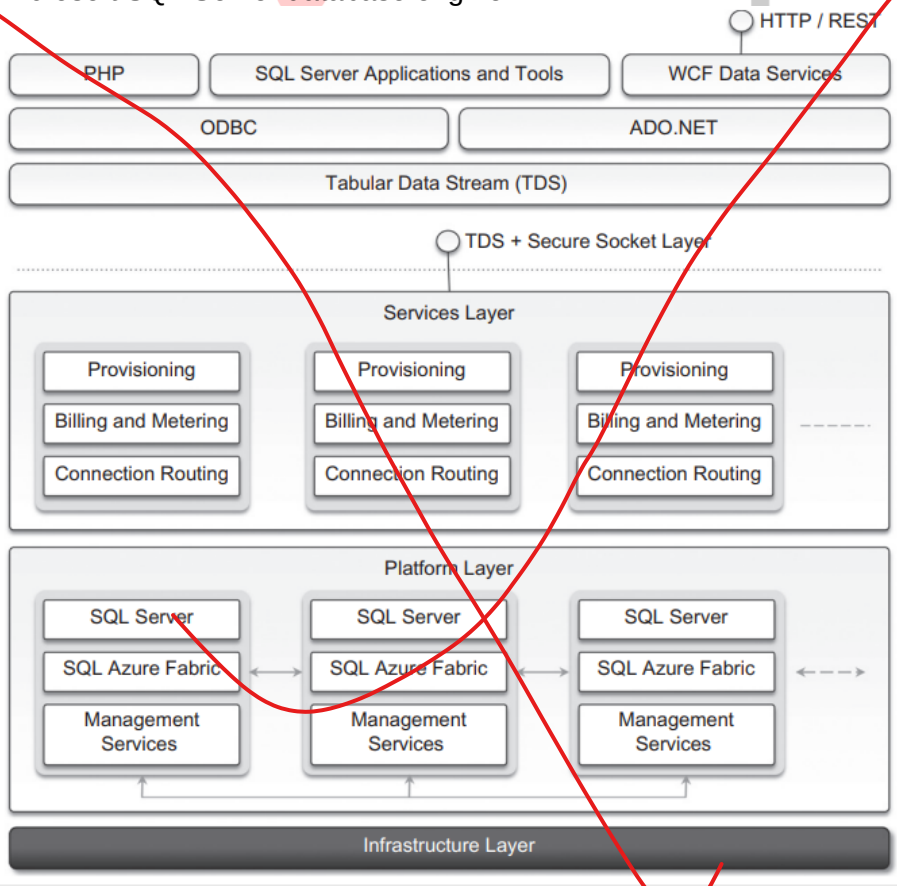
1. Windows Azure Connect: Allows easy setup of IP-based network connectivity among machines hosted on the private premises and the roles deployed on the Azure Cloud.
2. Windows Azure Traffic Manager: Provides load-balancing features for services listening to the HTTP or HTTPS ports and hosted on multiple roles.

Windows Azure content delivery network

1. Windows Azure Content Delivery Network (CDN):- is the content delivery network solution that improves the content delivery capabilities of Windows Azure Storage and several other Microsoft services, such as Microsoft Windows Update and Bing maps.

SQL Azure

- SQL Azure is a cloud-based relational database service that is based on the Microsoft SQL Server database engine.



SQL Azure Architecture

- It uses the Tabular Data Stream (TDS) protocol to communicate with applications.
- The TDS protocol is a binary protocol that is used to send and receive data between SQL Server and client applications.
- It is a highly efficient protocol that is designed to handle large volumes of data.

- On the SQL Azure side, access to data is mediated by the service layer.
- The service layer is responsible for provisioning, billing, and connection-routing services.
- It also provides a number of other features, such as high availability, disaster recovery, and security.
- The service layer is logically part of server instances.
- Server instances are the basic units of computation in SQL Azure. They are
- managed by SQL Azure Fabric, which is the distributed database middleware that constitutes the infrastructure of SQL Azure.
- SQL Azure Fabric is deployed on Microsoft data centers. It is a highly scalable and reliable platform that can handle large volumes of data and traffic.
- When a client application connects to SQL Azure, it is first routed to the service layer. The service layer then authenticates the client application and assigns it to a server instance.
- The client application can then begin sending and receiving data from the server instance.
- SQL Azure Fabric is responsible for distributing data and traffic across the server instances.
- It also ensures that the server instances are highly available and that data is protected from loss or corruption.

Windows Azure platform appliance

- The Windows Azure platform can also be deployed as an appliance on third-party data centers and constitutes the cloud infrastructure governing the physical servers of the datacenter.
- The Windows Azure Platform Appliance includes Windows Azure, SQL Azure, and Microsoft specified configuration of network, storage, and server hardware.
- The appliance is a solution that targets governments and service providers who want to have their own cloud computing infrastructure.

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Unit V – Cloud Applications

Cloud applications

- Cloud computing has gained huge popularity in industry due to its ability to host applications for which the services can be delivered to consumers rapidly at minimal cost.
- Applications from a range of domains, from scientific to engineering, gaming, and social networking, are considered.

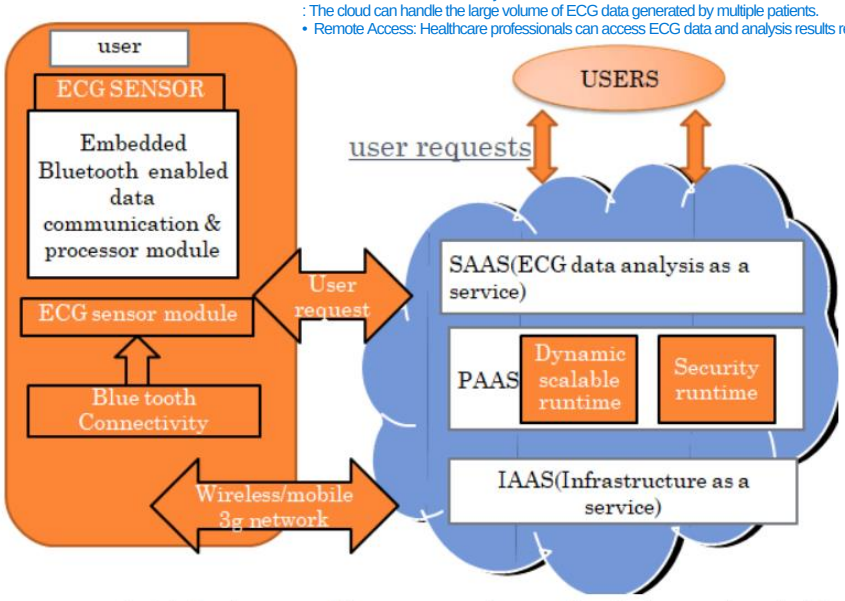
1. Scientific applications

- Scientific applications are a sector that is increasingly using cloud computing systems and technologies.
- Cloud computing systems meet the needs of different types of applications in the scientific domain: high-performance computing (HPC) applications, high-throughput computing (HTC) applications, and data-intensive applications.

a. Healthcare: ECG analysis in the cloud

- Healthcare is a domain in which computer technology has found several and diverse applications: from supporting the business functions to assisting scientists in developing solutions to cure diseases.
- An important application is the use of cloud technologies to support doctors in providing more effective diagnostic processes.
- In particular, here we discuss electrocardiogram (ECG) data analysis on the cloud
- An ECG, or electrocardiogram, is a recording of the electrical activity of the heart. It is a non-invasive test that is used to diagnose and monitor a variety of heart conditions, including arrhythmias, heart attacks, and heart failure.
- Cloud computing technologies allow the remote monitoring of a patient's heartbeat data, data analysis in minimal time, and the notification of first-aid personnel and doctors should study these data and reveal potentially dangerous conditions.
- An illustration of the infrastructure and model for supporting remote ECG monitoring is shown in below figure

Cloud computing allows for storing and processing large amounts of ECG data from multiple patients, enabling doctors and healthcare providers to access this information from anywhere.
: The cloud can handle the large volume of ECG data generated by multiple patients.
• Remote Access: Healthcare professionals can access ECG data and analysis results remotely, improving decision-making and response times



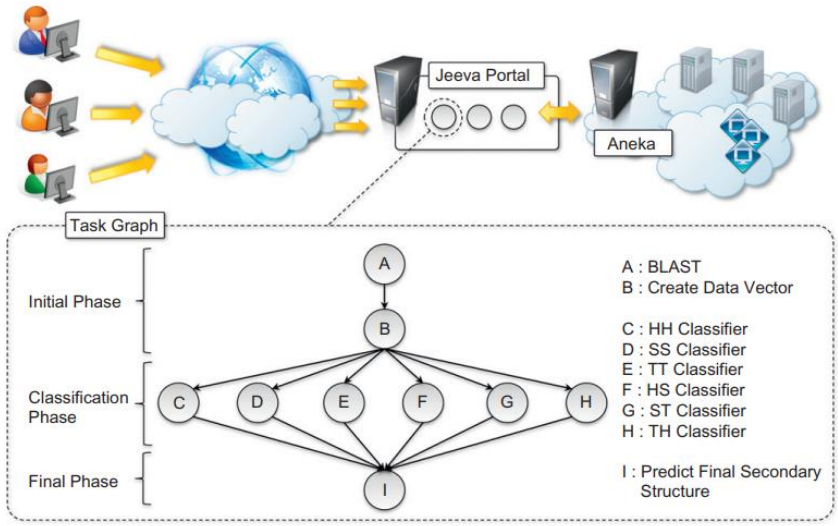
- Wearable computing devices equipped with ECG sensors constantly monitor the patient's heartbeat. Such information is transmitted to the patient's mobile device, which will eventually forward it to the cloud-hosted Web service for analysis.
- The Web service forms the front-end of a platform that is entirely hosted in the cloud and that leverages the three layers of the cloud computing stack: SaaS, PaaS, and IaaS.
- Set of operations involving the extraction of the waveform from the heartbeat data and the comparison of the waveform with a reference waveform to detect anomalies. If anomalies are found, doctors and first-aid personnel can be notified to act on a specific patient.

b. Biology: protein structure prediction

- Protein structure prediction is the process of determining the three-dimensional structure of a protein from its amino acid sequence, which is the primary structure.
- Protein structure prediction is a computationally intensive task that is fundamental to different types of research in the life sciences
- Protein structure prediction can help to understand the function, evolution, and regulation of proteins, as well as to design new drugs and biomolecules.
- One project that investigates the use of cloud technologies for protein structure prediction is Jeeva —an integrated Web portal that enables scientists to offload the prediction task to a computing cloud based on Aneka

Protein structure prediction requires massive computational power for simulations (like protein folding). Cloud platforms like AWS, Google Cloud, and Microsoft Azure provide the necessary resources without needing organizations to invest in expensive on-premise hardware.

Cloud-based services can use machine learning models, such as AlphaFold, to predict protein structures accurately. These services allow biologists to upload protein sequences and receive the predicted structures quickly, reducing the time and cost associated with traditional laboratory methods.



- Jeeva uses machine learning techniques, such as support vector machines, for determining the secondary structure of proteins.
- These techniques translate the problem into one of pattern recognition, where a sequence has to be classified into one of three possible classes (E, H, and C).
- A popular implementation based on support vector machines divides the pattern recognition problem into three phases: initialization, classification, and a final phase

- Jeeva can take advantage of parallel execution in the classification phase, where multiple classifiers are executed concurrently. This can reduce the computational time of the prediction.
- Jeeva submits the prediction task to Aneka as a task graph, and Aneka distributes and executes the task on the cloud. Jeeva then displays the results on the web portal.

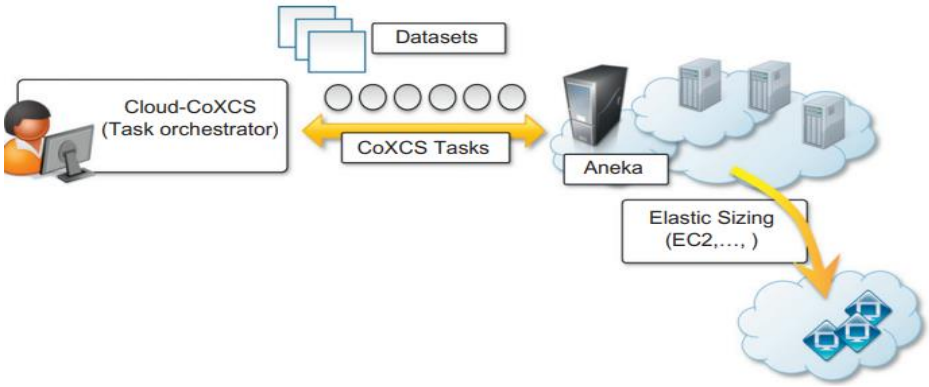
c. Biology: Gene expression for data analysis for cancer

: In cancer research, analyzing gene expression data helps in understanding which genes are upregulated or downregulated in cancer cells. This can be used for diagnosing cancer, determining its stage, and predicting patient outcomes.

Cloud computing enables storage of these large datasets and provides tools to analyze them.

- Gene expression profiling is a technique that measures the activity of thousands of genes at once, to create a global picture of cellular function.
- Gene expression profiling can help understand the biological processes that are triggered by medical treatment at a cellular level, and also identify the effects of a specific treatment on the genes that regulate cell growth and proliferation.
- Gene expression profiling can help in designing more effective and personalized drugs for cancer patients, by providing a more accurate classification of tumors based on their gene expression patterns.
- One of the challenges of gene expression profiling is the high dimensionality of the data, which means that there are many more genes than samples to analyze. This makes it difficult to find the most relevant genes for classification and prediction.
- One of the methods that can be used to overcome this challenge is learning classifier systems, or LCS, which are a paradigm of rule-based machine learning methods that combine a discovery component (e.g. typically a genetic algorithm) with a learning component (performing either supervised learning, reinforcement learning, or unsupervised learning).
- One of the examples of learning classifier systems that has been successfully applied to gene expression data is the eXtended Classifier System (XCS), which is an accuracy-based system that works by observing the environment, taking input from it and applying suitable actions.
- A variation of XCS, called CoXCS, has proven to be more effective in dealing with high dimensional datasets, such as microarray gene expression data sets. CoXCS divides the entire search space into subdomains and employs the standard XCS algorithm in each of these subdomains.

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- Cloud-CoXCS is a cloud-based implementation of CoXCS that leverages Aneka, a platform and framework for developing distributed applications on the Cloud.

Aneka provides a rich set of APIs and tools for developers to transparently exploit the resources of a heterogeneous network of desktop PCs and servers or datacenters on demand.

- Cloud-CoXCS uses Aneka to solve the classification problems in parallel and compose their outcomes. The algorithm is controlled by strategies, which define the way the outcomes are composed together and whether the process needs to be iterated.
- Cloud-CoXCS benefits from the advantages of the cloud computing model, such as lower costs, faster time to market, and easier maintenance and upgrades. Cloud-CoXCS also adapts to the dynamic nature of XCS, by using scalable middleware such as Aneka to adjust the number of required compute resources to execute it according to the load and performance of the system.

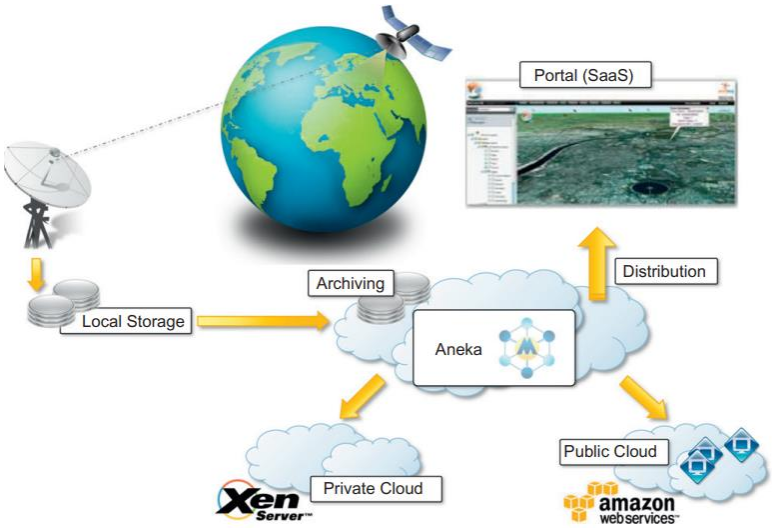
d. Geo science: satellite image processing

• Satellite image processing involves the analysis of images captured by satellites to monitor and study the Earth's surface. It's used for various purposes, including climate change monitoring, disaster management, environmental monitoring, and resource exploration

- Geoscience applications are programs that deal with data about the earth and its features, such as land, water, plants, and animals.
- Geoscience applications use a lot of data that comes from different sources, such as sensors and satellites, that measure and observe the earth.
- Geoscience applications need a lot of computing power to process and analyze the data, and to make maps and models that show the results.
- Cloud computing is a technology that allows users to access computing resources over the internet, without having to own or manage them. Cloud computing can provide more computing power, storage, and flexibility for geoscience applications.
- One example of a cloud-based geoscience application is a system that processes satellite images of the earth. Satellite images are pictures taken by satellites that orbit the earth. They can show different features and changes of the earth, such as weather, vegetation, land use, and disasters.
- The system uses a cloud platform called Aneka, which controls the importing of data and the execution of image-processing tasks. Aneka can also adjust the number of resources needed according to the workload and the performance of the system.

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- The system also provides a web application that offers different services for users, such as generating codes and visualizing data. Users can access the web application from anywhere and from any device that is connected to the internet.
- The system shows how cloud computing can help geoscience applications to handle large amounts of data and to produce useful information for decision makers.



Business and Consumer application

The business and consumer sector is the one that probably benefits the most from cloud computing technologies.

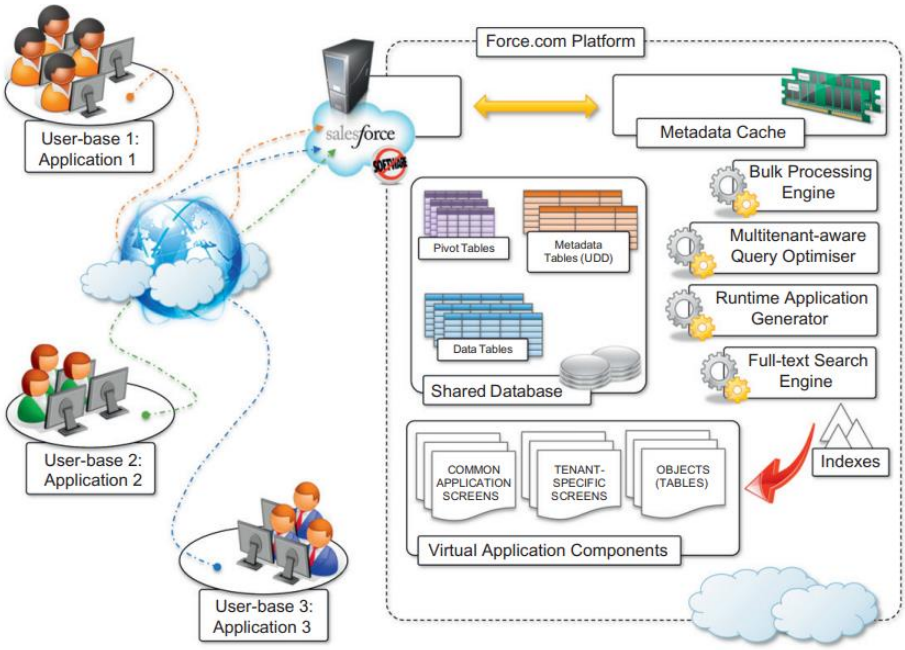
CRM is a system used by businesses to manage interactions with current and potential customers. It helps organize customer data, track sales, improve customer service, and analyze data to make informed decisions.
Centralized Data Management: Cloud-based CRM platforms store all customer information in a centralized system that is accessible from anywhere via the internet.
Scalability: Businesses can scale the CRM system up or down based on their needs without worrying about infrastructure constraints.

1. **CRM and ERP**
   - CRM stands for Customer Relationship Management. It refers to a set of practices, strategies, and technologies that companies use to manage and analyze customer interactions and data throughout the customer lifecycle. The goal of CRM is to improve relationships with customers, streamline processes, and ultimately increase profitability.
   - ERP stands for enterprise resource planning. It is a software system that integrates various business processes, such as accounting, human resources, supply chain management, and manufacturing. ERP systems help organizations to manage their resources more efficiently and to make better decisions.
   - Cloud CRM and cloud ERP applications are software that run on the cloud, which means that they are accessed over the internet and hosted by a third-party provider. This way, businesses do not have to install, maintain, or upgrade the software on their own servers, and they can pay for the services they use on a subscription basis .
   - Example for CRM :- Salesforce.com, Microsoft dynamics CRM, Zoho CRM etc..
   - Example for ERP: - NetSuite ERP, SAP ERP, Oracle ERP cloud etc…

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   a. **Salesforce.com**
      - Salesforce.com is a software that helps you manage your relationships and interactions with your customers and prospects, and improve your sales and marketing performance.

It is designed to help businesses manage customer relationships, improve sales, and automate marketing.



Salesforce.com and Force.com architecture

- Salesforce.com provides customizable CRM solutions that can be integrated with additional features developed by third parties.
- Salesforce.com is built on the Force.com cloud development platform, which is a scalable and high-performance middleware that executes all the operations of Salesforce.com applications.
- Force.com has a metadata architecture that stores both the application structure and the application data in a common store.
- A runtime engine executes the application logic by retrieving its metadata and performing the operations on the data.
- Force.com provides different ways for users to customize and extend their applications, either by using the native Force.com application framework or by using programmatic APIs in the most popular programming languages.
- Force.com also provides a scripting language called APEX, which is a Java-like language that provides object-oriented and procedural capabilities for defining scripts or triggers. APEX also allows users to express searches and queries to access the data managed by the Force.com platform.

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b. Microsoft dynamics CRM

Microsoft Dynamics CRM (now part of Dynamics 365) is a cloud-based CRM platform tightly integrated with Microsoft's ecosystem

- Microsoft Dynamics CRM is a software that helps you manage your relationships and interactions with your customers and prospects, and improve your business performance and efficiency
- Microsoft Dynamics CRM is available either as an on-premise installation or as an online service that is hosted by Microsoft in its datacenters across the world
- Offers a 99.9% Service Level Agreement (SLA) to customers, with bonus credits if the system fails to meet the agreed-upon service levels
- Each CRM instance is deployed on a separate database, contributing to data isolation and security.
- Dynamics CRM provides functionalities for marketing, sales, and advanced customer relationship management.
- These features contribute to managing various aspects of customer interactions and relationships.

- Microsoft Dynamics CRM can be accessed through a web browser interface or programmatically by using SOAP and RESTful web services.
- Microsoft Dynamics CRM can be extended and customized by developing plug-ins that allow you to implement specific behaviors triggered by certain events, such as creating, updating, or deleting records. You can also leverage the power of Windows Azure, Microsoft's cloud computing platform, to develop and integrate new features and functionalities for your Microsoft Dynamics CRM application.

c. NetSuite NetSuite, a product by Oracle, is an all-in-one cloud-based solution that combines ERP, CRM, eCommerce, and financial management. It's particularly popular among growing businesses for its integrated approach.

- NetSuite provides a collection of applications to manage various aspects of business enterprises.
- The offerings are divided into three major products: NetSuite Global ERP, NetSuite Global CRM, and NetSuite Global Ecommerce.
- NetSuite One World integrates all three major products (ERP, CRM, and Ecommerce) into an all-in-one solution, providing a comprehensive business management platform.
- NetSuite's services are powered by two large datacenters on the East and West coasts of the United States, connected by redundant links.
- The infrastructure allows NetSuite to guarantee 99.5% uptime to its customers.
- NetSuite offers an infrastructure and a development environment for implementing customized applications.
- NetSuite Business Operating System (NS-BOS) is a stack of technologies for building SaaS (Software as a Service) business applications that leverage the capabilities of NetSuite products.
- The NetSuite Business Suite components offer accounting, ERP, CRM, and ecommerce capabilities as part of their online development environment.
- SuiteFlex is an online development environment provided by NetSuite, allowing the integration of accounting, ERP, CRM, and ecommerce capabilities into new web applications.
- SuiteBundler is a tool for packaging these applications for distribution.
- The entire infrastructure, including NetSuite Business Operating System components, is hosted in NetSuite datacenters.
- The datacenters provide warranties regarding application uptime and availability.

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Productivity

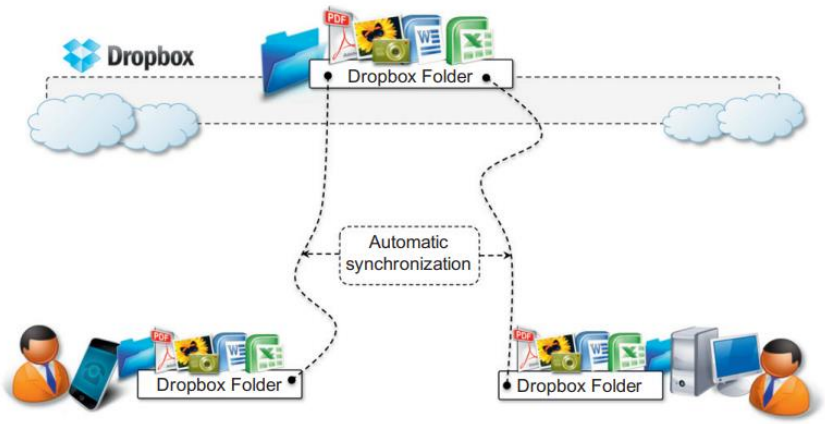
- Productivity applications replicate in the cloud some of the most common tasks that we are used to performing on our desktop: from document storage to office automation and complete desktop environments hosted in the cloud

a. Dropbox and iCloud

- Online document storage is one of the applications of cloud computing that allows users to store, synchronize, and share their files across different platforms and devices.
- Some examples of online document storage solutions, such as Dropbox, iCloud, Windows Live, Amazon Cloud Drive, and CloudMe

i. Dropbox

- Dropbox is highlighted as one of the most popular solutions for online document storage.
- It allows users to synchronize files across different platforms and devices seamlessly.
- Users can access their Dropbox folder through a browser or by installing a Dropbox client. Modifications in the folder are synchronized across devices in real-time.
- Dropbox is available on various platforms, including Windows, Mac, Linux, and mobile devices.



ii. iCloud

- iCloud is mentioned as a cloud-based document-sharing application provided by Apple.
- It is designed for transparent synchronization of iOS-based devices.
- iCloud automatically syncs documents, photos, and videos as changes are made, without requiring explicit operations.
- The system automates common operations seamlessly, such as automatically making a picture taken on an iPhone available in iPhoto on a Mac.
- iCloud's capability is limited to iOS devices, and there are no plans mentioned to provide a web-based interface for broader platform accessibility.

b. Google docs

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- Google Docs is a software as a service (SaaS) application that delivers the basic office automation capabilities with support for collaborative editing over the Web.
- Google Docs runs on top of the Google distributed computing infrastructure, which allows the system to dynamically scale according to the number of users using the service.
- Do not need to install or maintain any software on your own devices, and can access files from anywhere and from any device that is connected to the Internet.
- Google Docs allows, to create and edit text documents, spreadsheets, presentations, forms, and drawings, using various formatting tools, templates, and add-ons to enhance work.
- We can also import and export files in different formats, such as Microsoft Office, OpenOffice, PDF, and HTML.
- Google Docs supports collaborative editing over the Web for most of the applications included in the suite.
- Google Docs also allows to work offline if Internet connectivity is not available. Enable offline access for your files and edit them without an Internet connection. Changes will be automatically synced when you reconnect to the Internet.

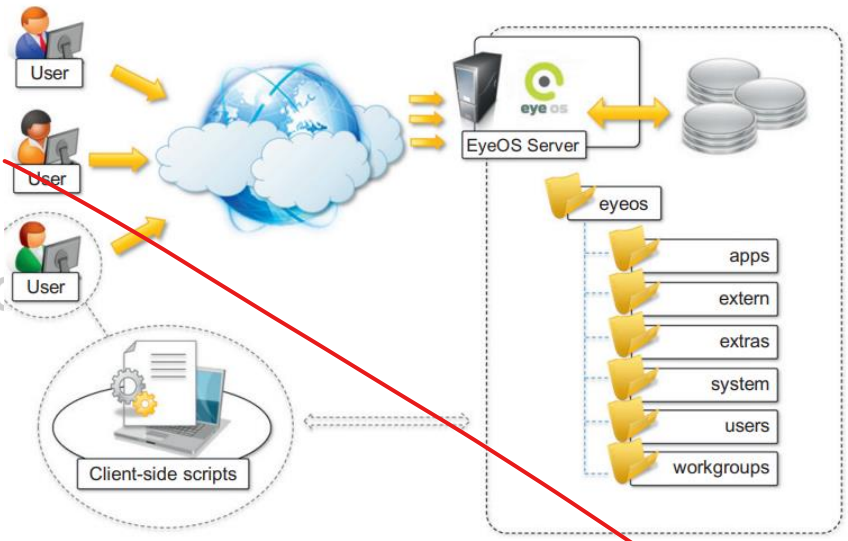
Slack:
Purpose: A communication platform for team collaboration.
Features:
Organized messaging through channels.
Integration with other productivity tools like Google Drive and Trello.
File sharing and video calls.
Use Case: A software development team uses Slack to streamline communication and coordinate tasks across multiple projects.

c. Cloud desktops: EyeOS and XIOS/3

- AJAX stands for Asynchronous JavaScript and XML, which is a set of web development techniques that allow web applications to communicate with the server without reloading the page.
- AJAX enables web applications to be more interactive, responsive, and user-friendly.
- Cloud desktops are web applications that replicate the functionalities of a classic desktop environment, such as file and document management, and make them available through the web browser.
- Cloud desktops are powered by cloud computing, which delivers computing resources over the internet on demand.
- Examples of cloud desktops, such as EyeOS, XIOS/3, and CloudMe.

i. EyeOS

- EyeOS is a web desktop solution that replicates the functionalities of a classic desktop environment and comes with pre-installed applications for the most common file and document management tasks.
- EyeOS allows users to access their desktop environment from anywhere and from any internet-connected device, and also provides APIs for developing new applications and integrating new capabilities into the system.
- EyeOS stores the data about users and applications on the server file system, and renders the desktop environment in the client's browser by downloading all the JavaScript libraries required to build the user interface and implement the core functionalities of EyeOS.
- Each application loaded in the environment communicates with the server by using AJAX.



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ii. XIOS/3

- XIOS/3 is a web desktop environment that is part of the CloudMe application, which is a solution for cloud document storage.
- XIOS/3 leverages XML to implement many of the tasks of the OS, such as rendering user interfaces, defining application business logics, structuring file system organization, and even application development.
- XIOS/3 concentrates most of the functionalities on the client side, while implementing server-based functionalities by means of XML web services.

- The client side renders the user interface, orchestrates processes, and provides data-binding capabilities on XML data that is exchanged with web services.
- The server is responsible for implementing core functions such as transaction management and core logic of installed applications.
- XIOS/3 also provides an environment for developing applications (XIDE), which allows users to quickly develop complex applications by visual tools for the user interface and XML documents for business logic.

Social Networking

- Social networking applications have grown considerably in the last few years to become the most active sites on the Web.
- To sustain their traffic and serve millions of users seamlessly, services such as Twitter and Facebook have leveraged cloud computing technologies.
- The possibility of continuously adding capacity while systems are running is the most attractive feature for social networks, which constantly increase their user base.

a. Facebook

- Facebook is a social networking website that has more than 800 million users and offers various services and features to them.
- Facebook runs on two data centers that are built and optimized to reduce costs and environmental impact. The data centers use inexpensive hardware and customized open-source technologies.
- Facebook's platform is based on LAMP (Linux, Apache, MySQL, and PHP), which is a collection of technologies for developing web applications. Facebook also uses other services and tools that are developed in-house and in different languages, such as Thrift, Scribe, and others.
- Facebook's platform supports the creation and integration of third-party applications, such as social games and quizzes, with Facebook's core infrastructure and data.
- Facebook's platform uses the concept of social graph, which is a collection of interlinked information that is relevant for a given user. The social graph is composed by querying a distributed cluster of MySQL instances and using other services that are located closer to the data.

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Media applications

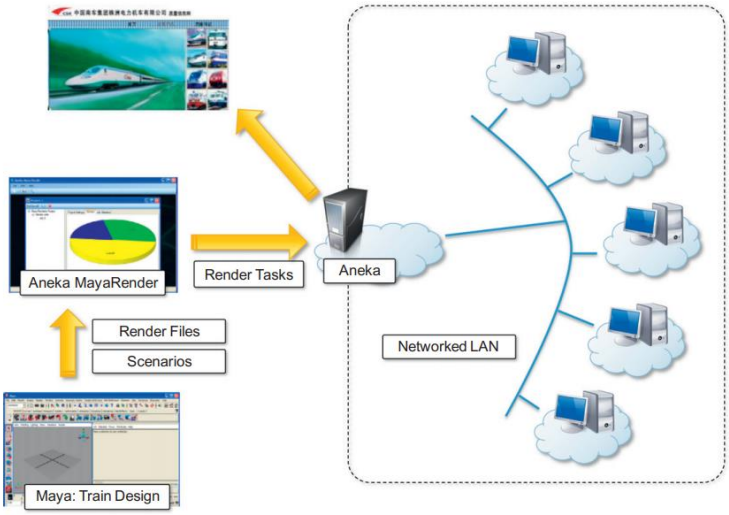
a. Animoto

- Animoto, a cloud-based media application that allows to create videos out of images, music, and video fragments.
- Animoto allows to create and edit videos online, by uploading your photos and videos and ordering them in the sequence you want them to appear, and selecting the song and the theme for the music and the effects.
- We can also choose from different templates and styles, and add text and captions to your videos.
- Animoto supports collaborative editing, which means that, you can share your videos with others and work on them together in real time.
- Animoto runs on top of the Amazon Web Services infrastructure, which allows the system to dynamically scale according to the number of users using the service.
- Animoto uses Amazon EC2 for the web front-end and the worker nodes, Amazon S3 for the storage of pictures, music, and videos, and Amazon SQS for connecting all the components.

- The system's auto-scaling capabilities are managed by Rightscale, which monitors the load and controls the creation and reclaim of new worker instances.

b. Maya rendering with aneka

- Rendering is the process of generating an image from a 3D model by simulating the effects of light and other physical phenomena.
- Rendering is computationally intensive and time-consuming, especially for high-quality images and animations.



- GoFront Group's engineering department implemented a private cloud solution for rendering train designs.
- GoFront is responsible for designing models of high-speed electric locomotives, metro cars,
- GoFront uses a specialized client interface to enter the details of the rendering process, such as the number of frames, the number of cameras, and other parameters. The client interface submits the rendering tasks to the Aneka Cloud, which distributes the load across all the available machines in the local network.
- Each rendering task triggers the execution of the local Maya batch renderer.
- The results of the rendering tasks are collected by the Aneka Cloud.
- By utilizing the private cloud during off-peak hours, GoFront reduced rendering time from days to hours.

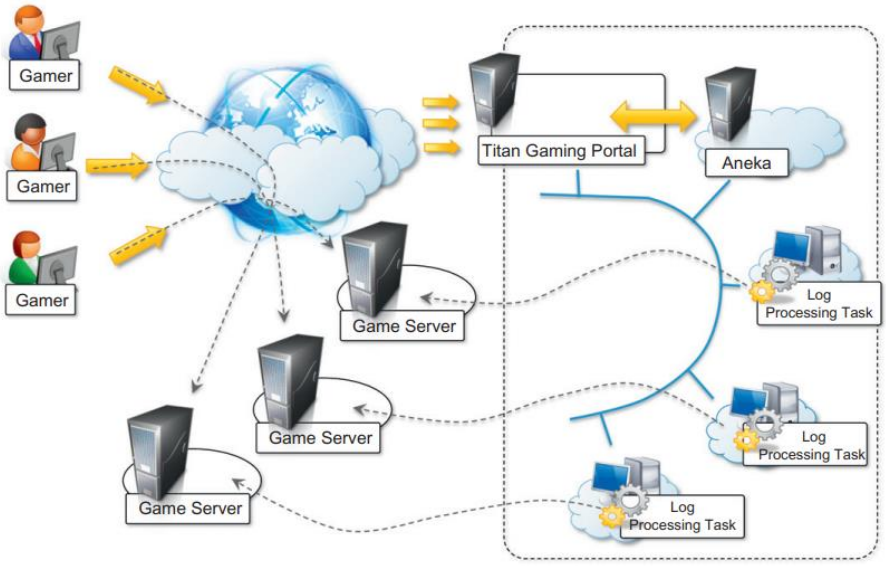
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c. Video encoding on the cloud. Encoding.com

- Video encoding and transcoding are techniques to convert video from one format to another, such as to make it more suitable for streaming, storage, or playback on different devices.
- Video encoding and transcoding can be computationally intensive and require a lot of storage space, especially for high-quality or large-scale videos.
- Encoding.com is an example of a software solution that offers video-transcoding services on demand and leverages cloud technology to provide both the computing power and the storage for video conversion.
- The service integrates with both Amazon Web Services and Rackspace, which are two of the leading cloud providers in the market.
- Users can access the service through various interfaces, such as the Encoding.com website, web service XML APIs, desktop applications, and watched folders. The service allows users to specify the source and destination formats of the video, as well as other editing options, such as thumbnails, watermarks, or logos.

Multiplayer Online gaming

- Online multiplayer gaming attracts millions of gamers around the world who share a common experience by playing together in a virtual environment that extends beyond the boundaries of a normal LAN.
- Online games support hundreds of players in the same session, made possible by the specific architecture used to forward interactions, which is based on game log processing.
- Game log processing is the technique of analyzing the data generated by the actions and interactions of the players in a game session.
- Game log processing can be used for various purposes, such as updating the game state, ranking the players, building statistics, etc.
- Game log processing can be computationally intensive and depend on the number of players and games. Moreover, gaming portals can face unpredictable and variable workloads that require flexible and scalable resources.



- Titan Inc. (now Xfire) is a company that provides a gaming portal for online multiplayer gaming.
- Titan Inc. used Aneka to implement a cloud-based game log processing system that can handle multiple logs and support more users.

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- The system uses a private cloud deployment that allows Titan Inc. to process the logs concurrently and efficiently.
- The system also uses a task graph to submit the log processing tasks to Aneka, and Aneka distributes and executes the tasks on the cloud.
- The system then displays the results on the gaming portal.